

Muscular Tissue

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Aims

Summarize the main similarities
& differences between 3
different types of muscle



1- skeletal muscle.



2- cardiac Muscle.



3- smooth Muscle.

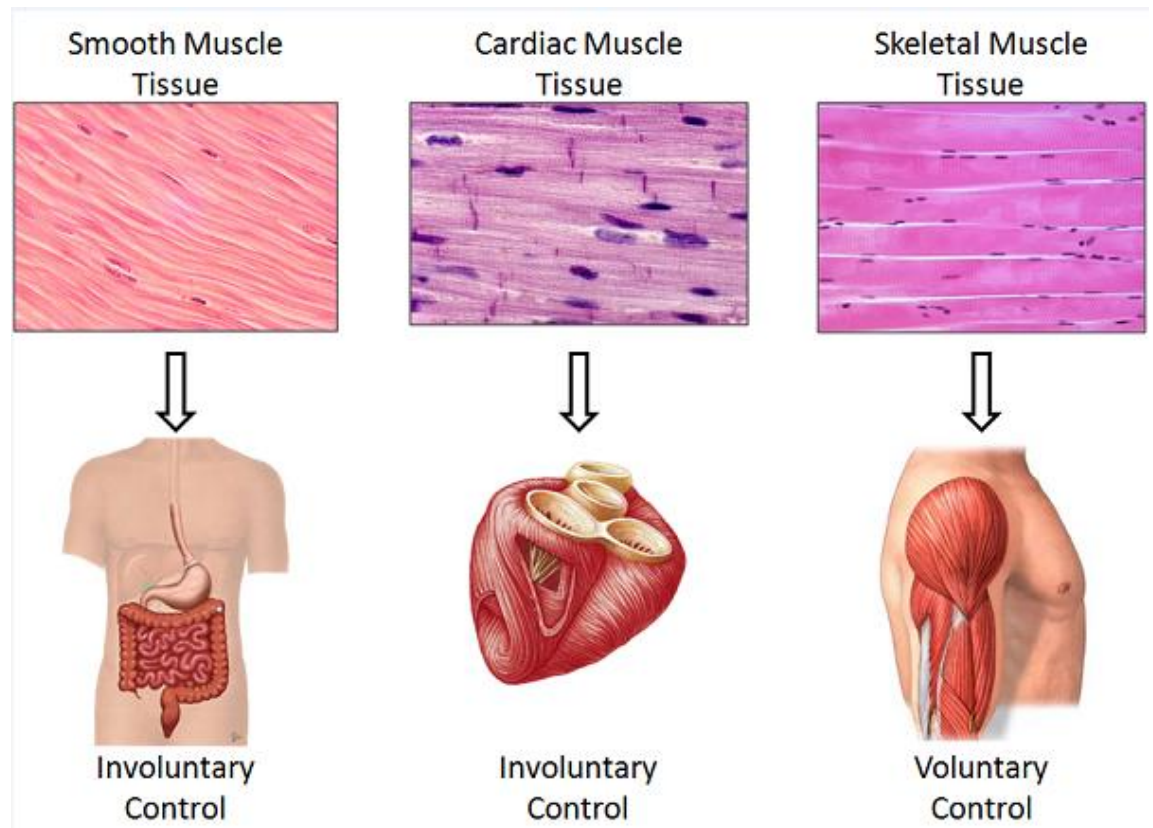


Each kind of muscle in the body
is made up of muscle cells

MUSCLE CHARACTERISTICS (VIP)

Features	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Striations	Yes	Yes	No
Fibers	Long, cylindrical, unbranched	Short, branched, anastomosing	Short, spindle shaped
Nuclei	Multiple, peripheral in cell	Single, central in cell	Single, central in cell
Cell junctions	No	Intercalated disks	Gap (nexus) junctions
T tubules	Well developed	Well developed	No
Sarcoplasmic reticulum	Highly developed; has terminal cisterns	Less well developed; small cisterns	Present, but poorly developed
Regeneration	Yes, satellite cells	No	Yes, mitosis
Contraction	Initiated by nerve action potential	Spontaneous; pacemaker system; modulated by nervous system and hormones	Spontaneous; modulated by nervous system and hormones
Main function	Voluntary movement of limbs, digits, face, tongue, and other muscles	Involuntary rhythmic contractions; pumps blood to muscles and organs; modulated by physiological and emotional factors	Involuntary control of blood vessel diameter, gut peristalsis, uterine contractions during childbirth, airway diameter, and others

Muscle types

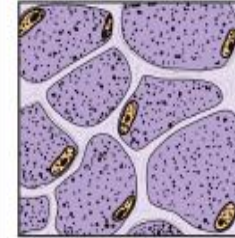


Muscle types

1- Skeletal muscle

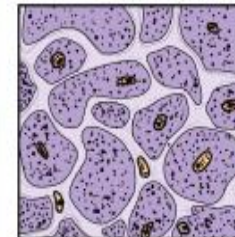


Cross sections



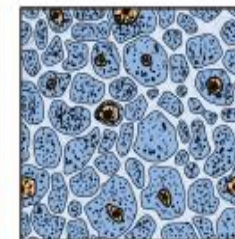
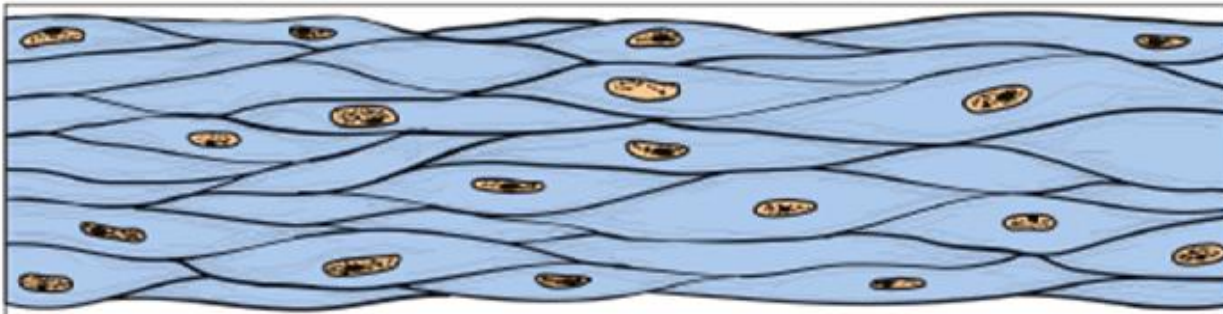
Strong, quick
discontinuous
voluntary
contraction

2- Cardiac muscle



Rhythmic
continuous
involuntary
contraction

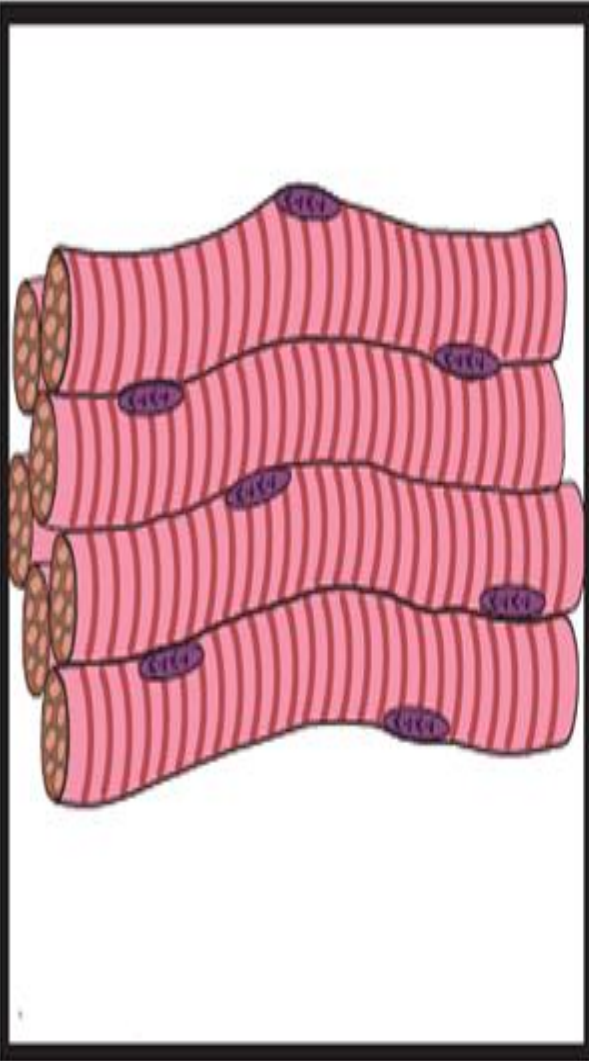
3-Smooth muscle



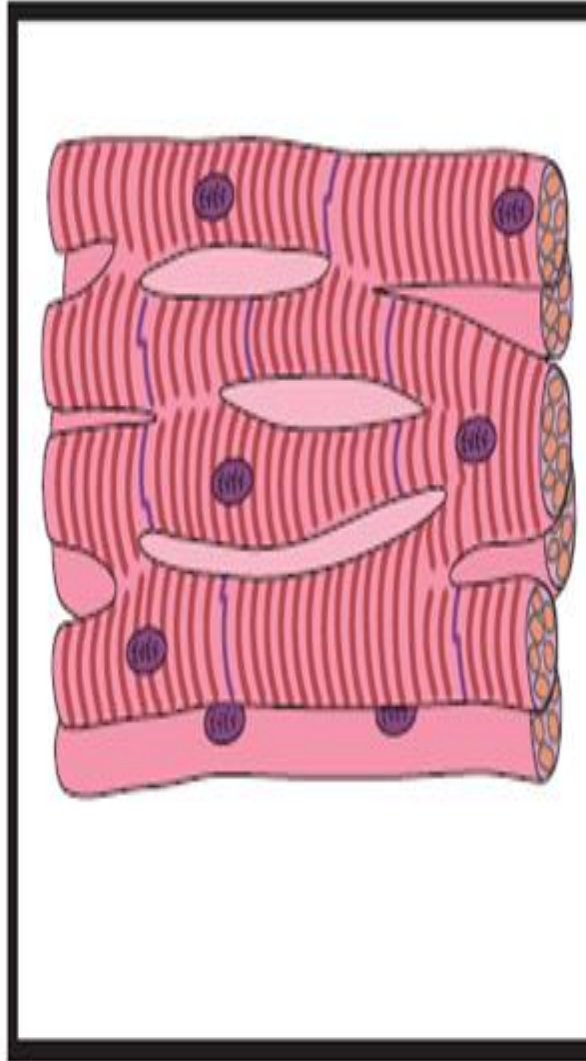
Weak, slow
involuntary
contraction

Muscle types

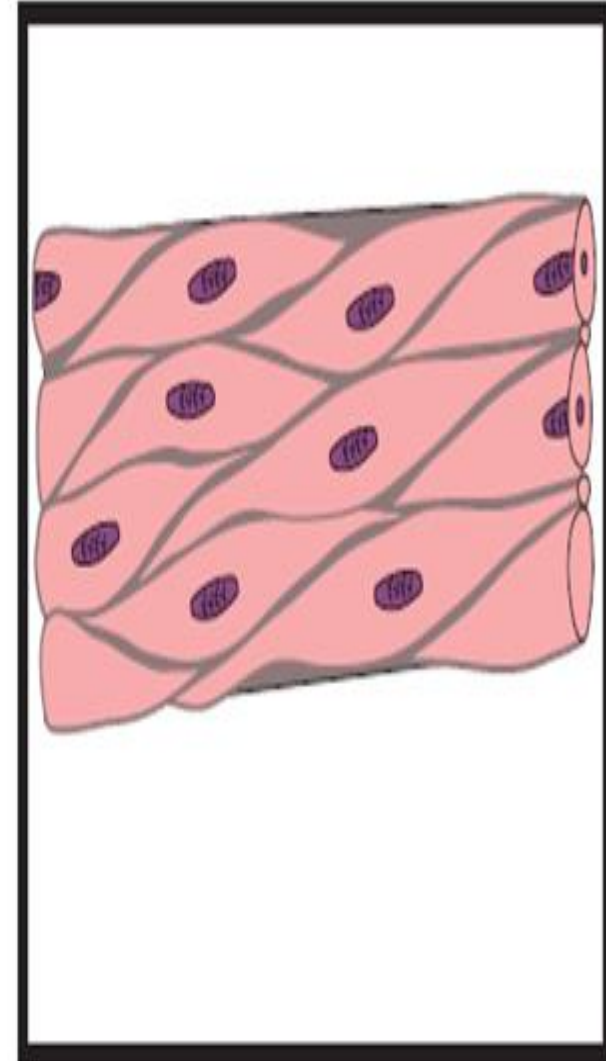
Skeletal muscle



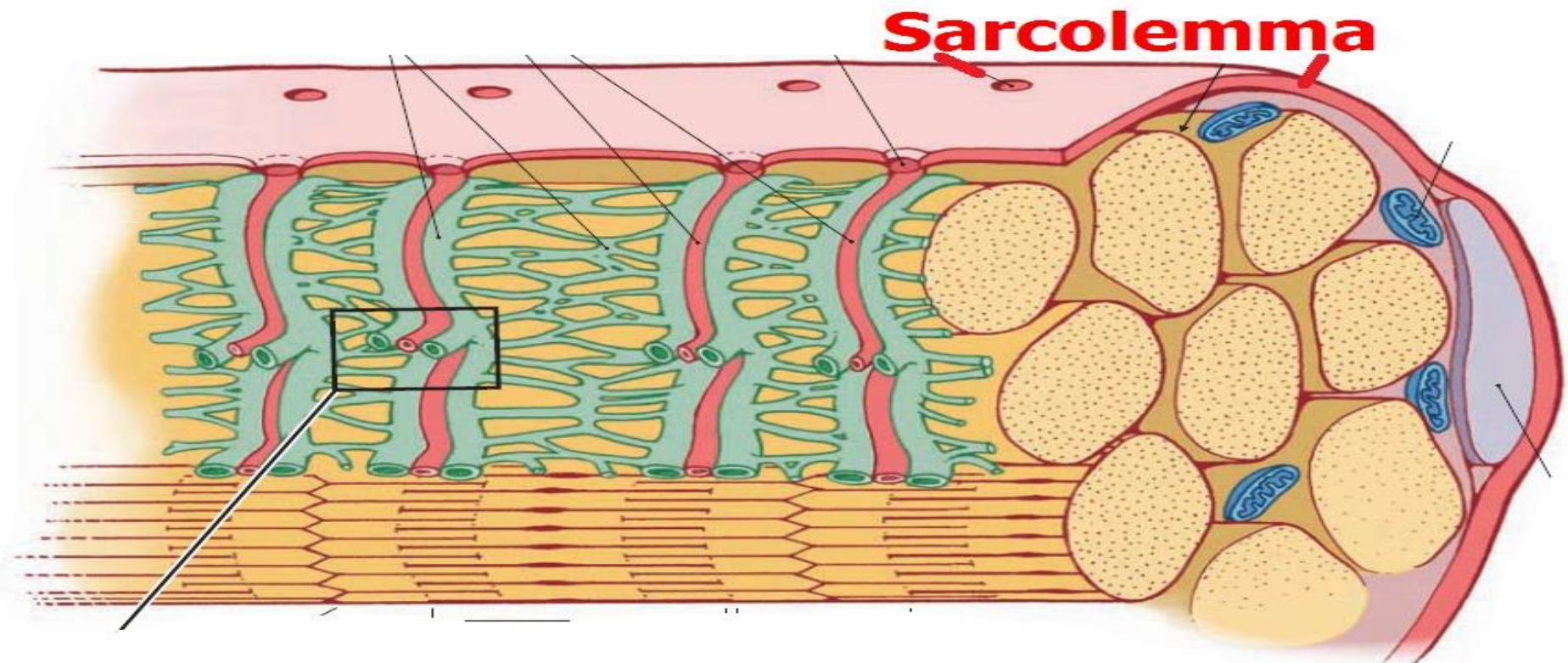
Cardiac muscle



Smooth muscle

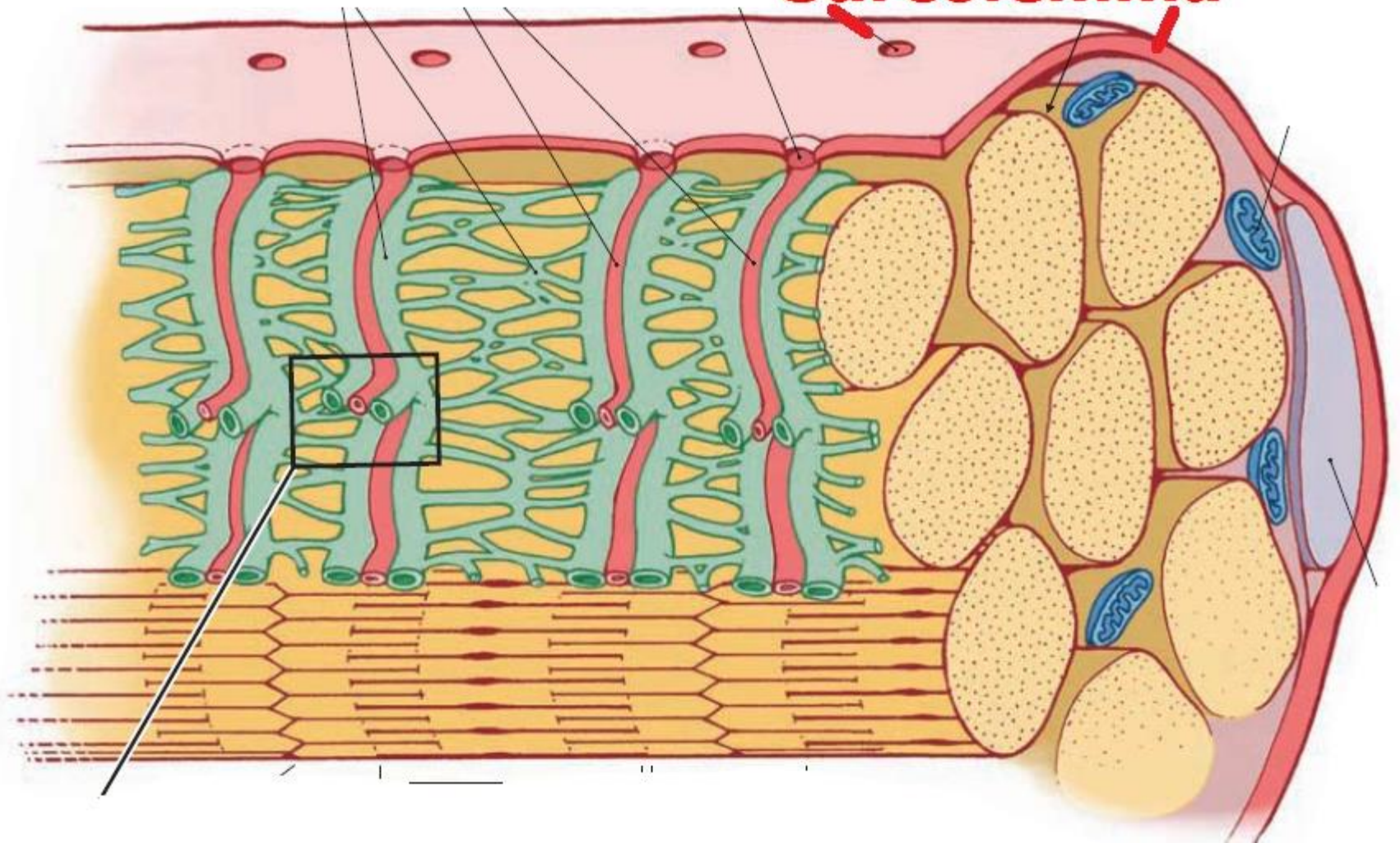


The organization of striated muscle fiber



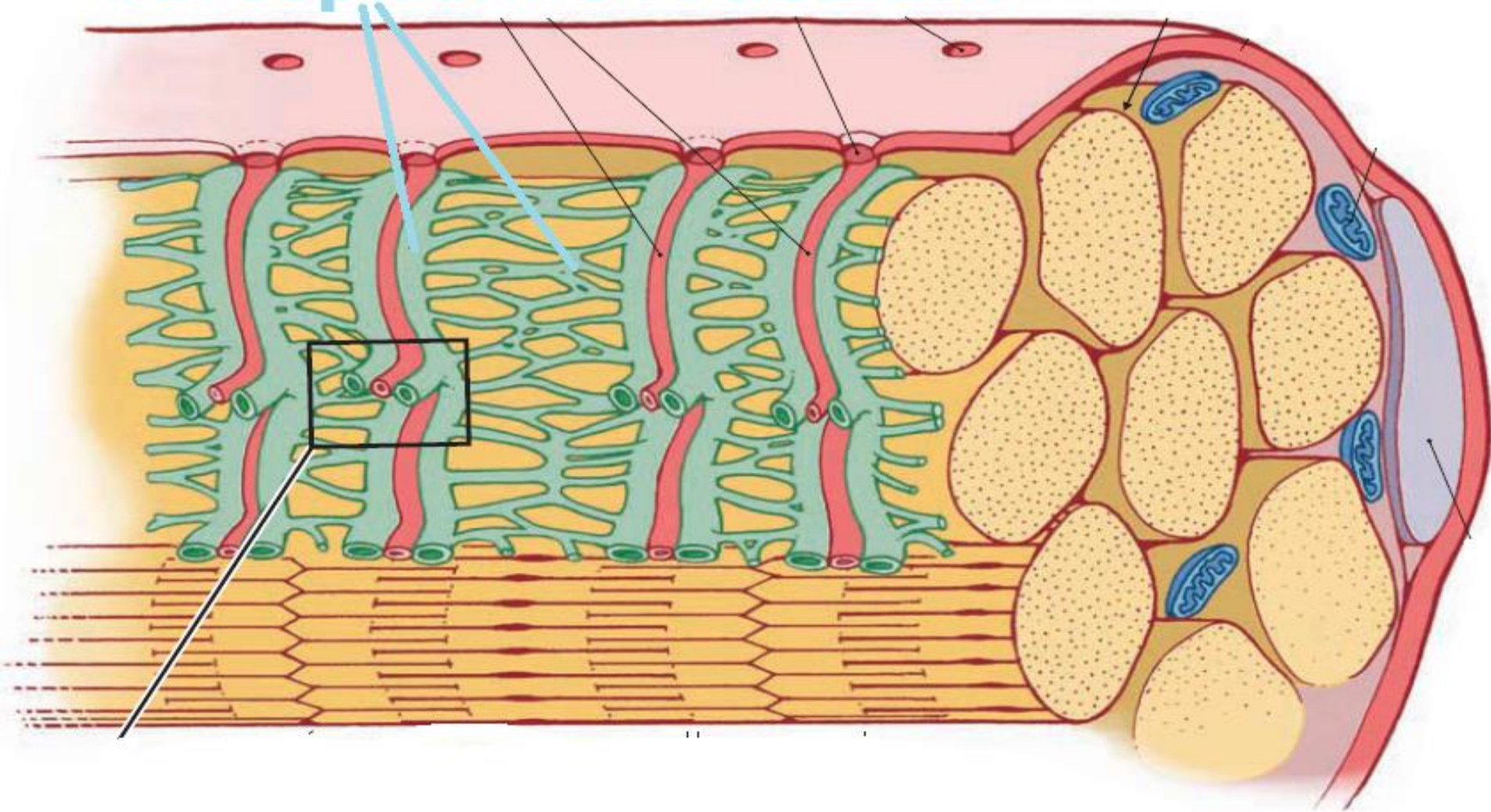
Sarcolemma is the cell membrane = plasmalemma.

Sarcolemma



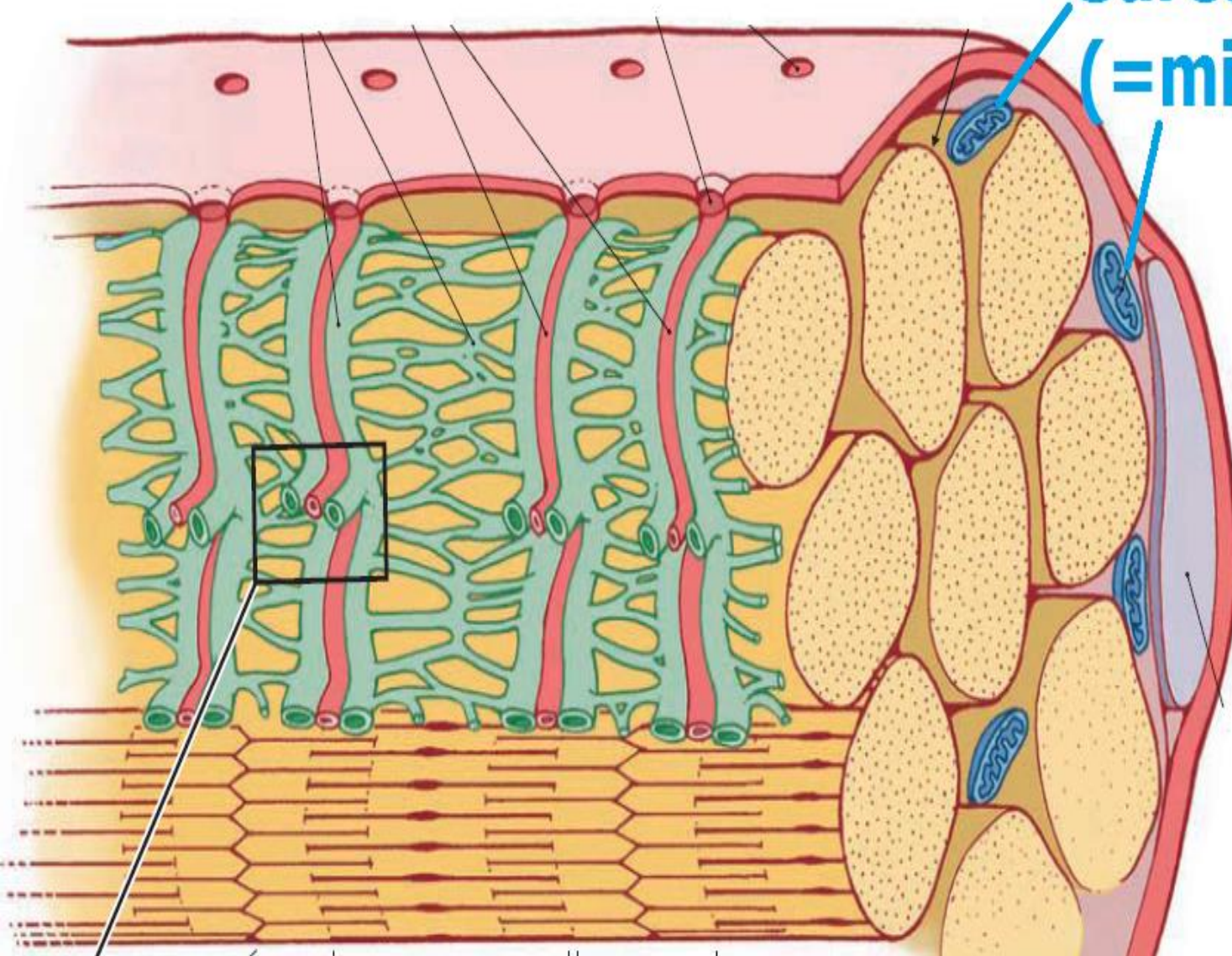
The **sar**co**plasm** is **cytoplasm** of muscle cells

Sarcoplasmic reticulum



Sarcoplasmic reticulum is (sER)smooth endoplasmic reticulum .

**Sarcosomes
(=mitochondria)**



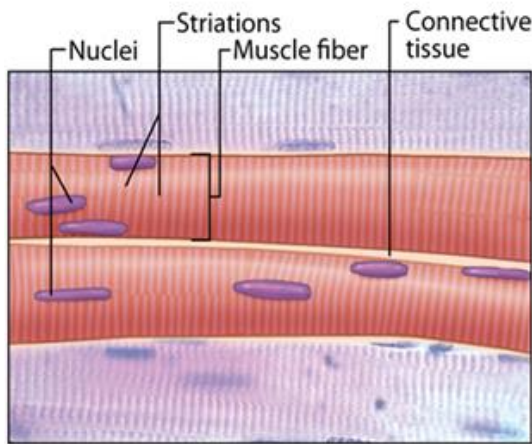
The sarcosomes are mitochondria.

Types of muscles

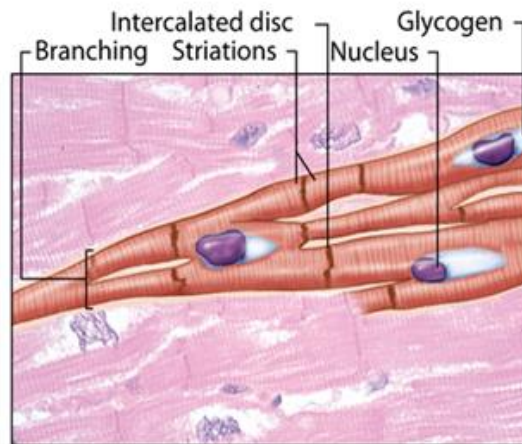
1 - Skeletal muscle

2- Cardiac muscle

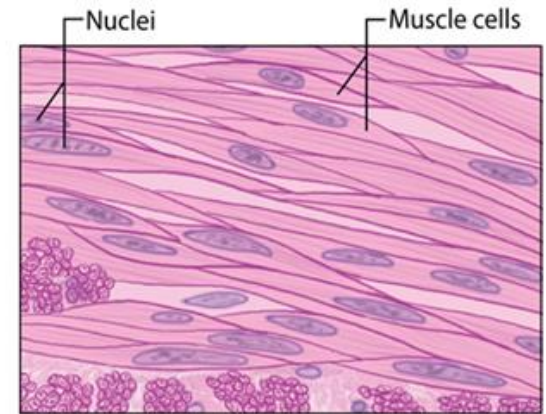
3-Smooth muscle



a Skeletal muscle

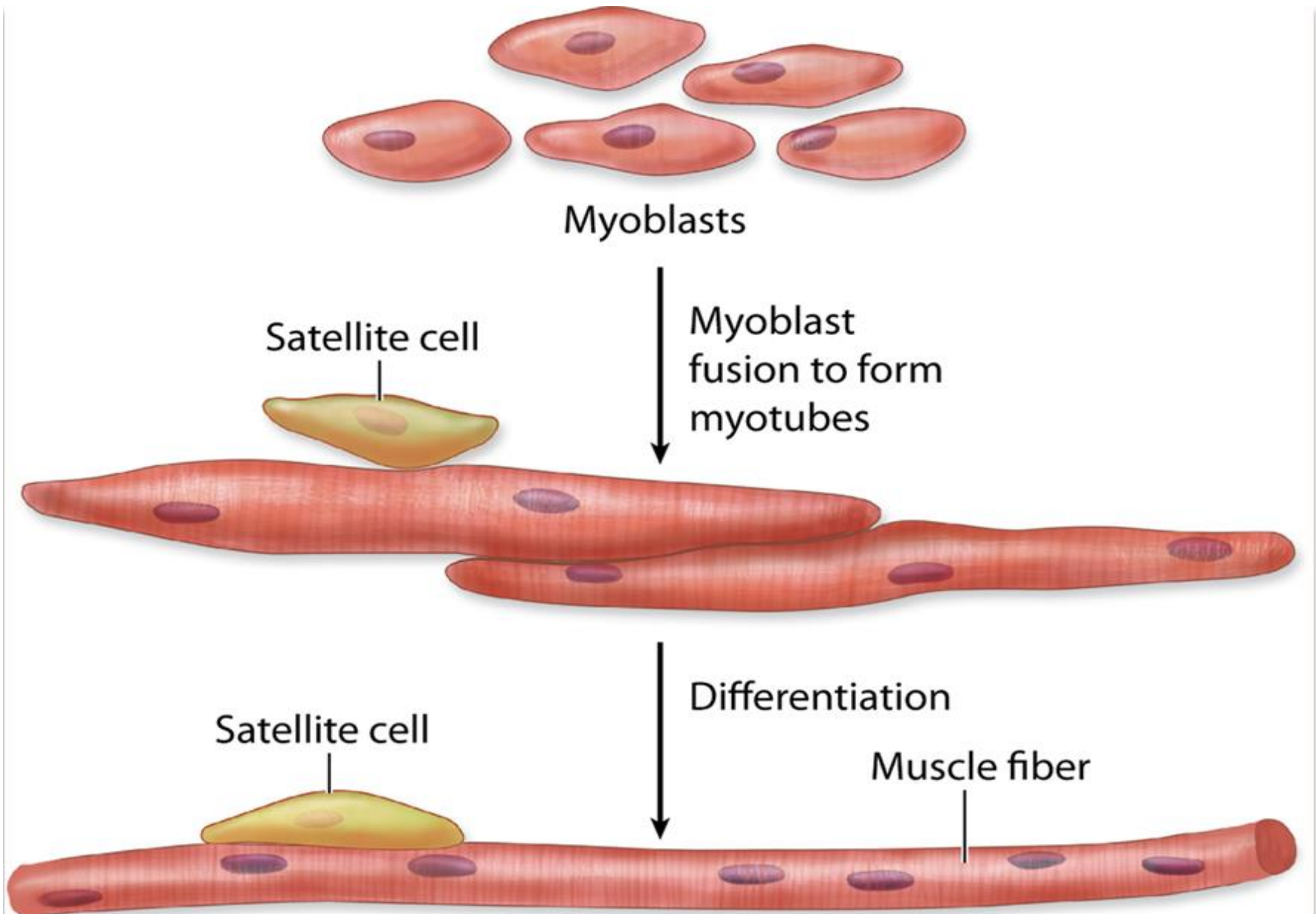


b Cardiac muscle

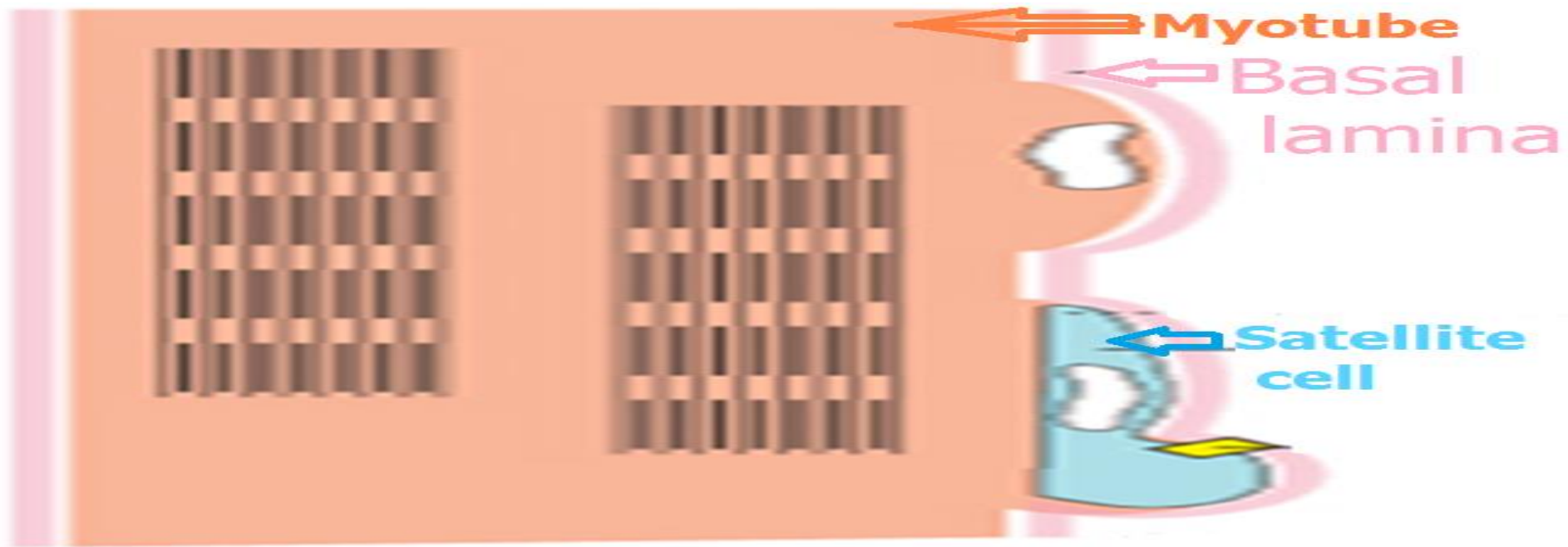


c Smooth muscle

Development of skeletal muscle

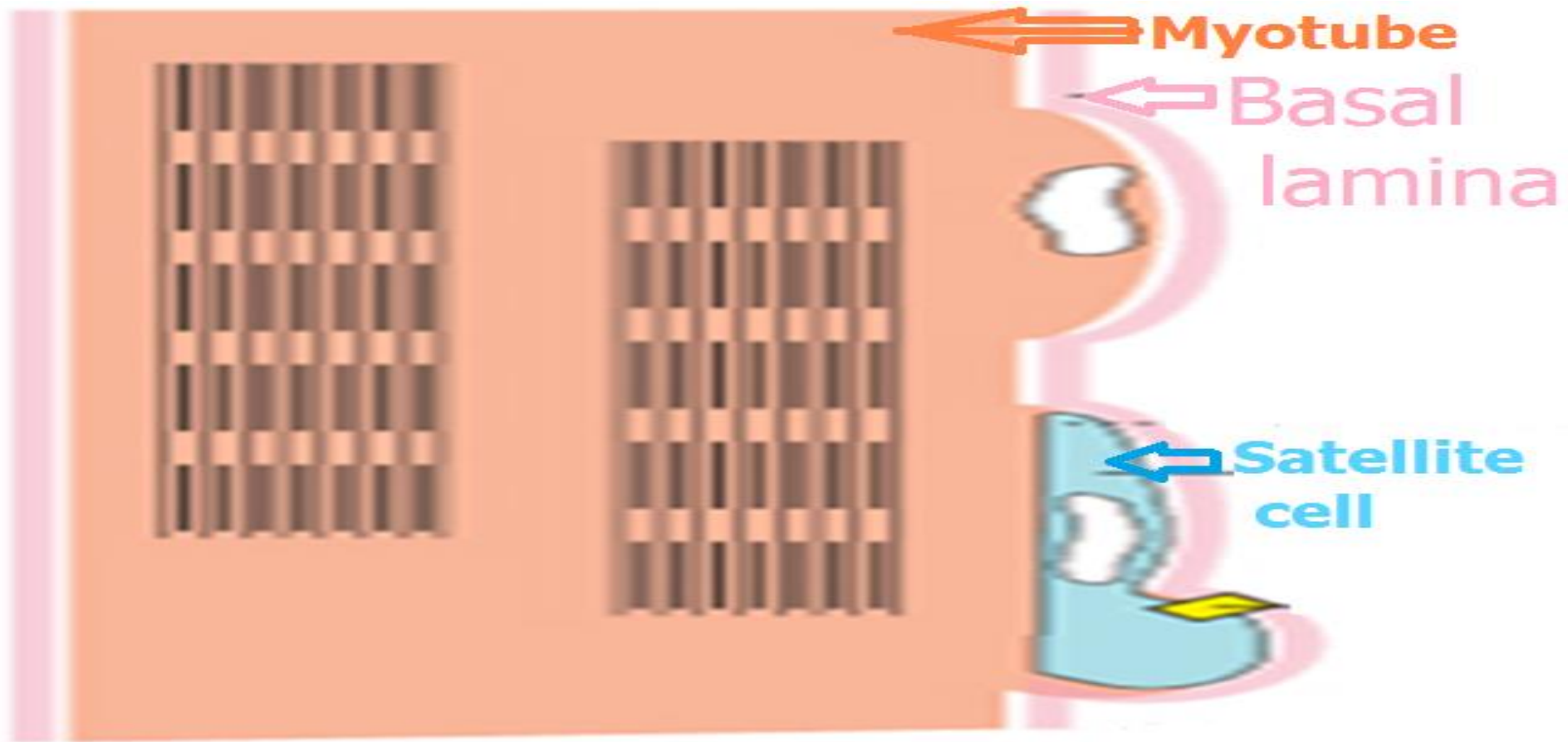


Satellite cells



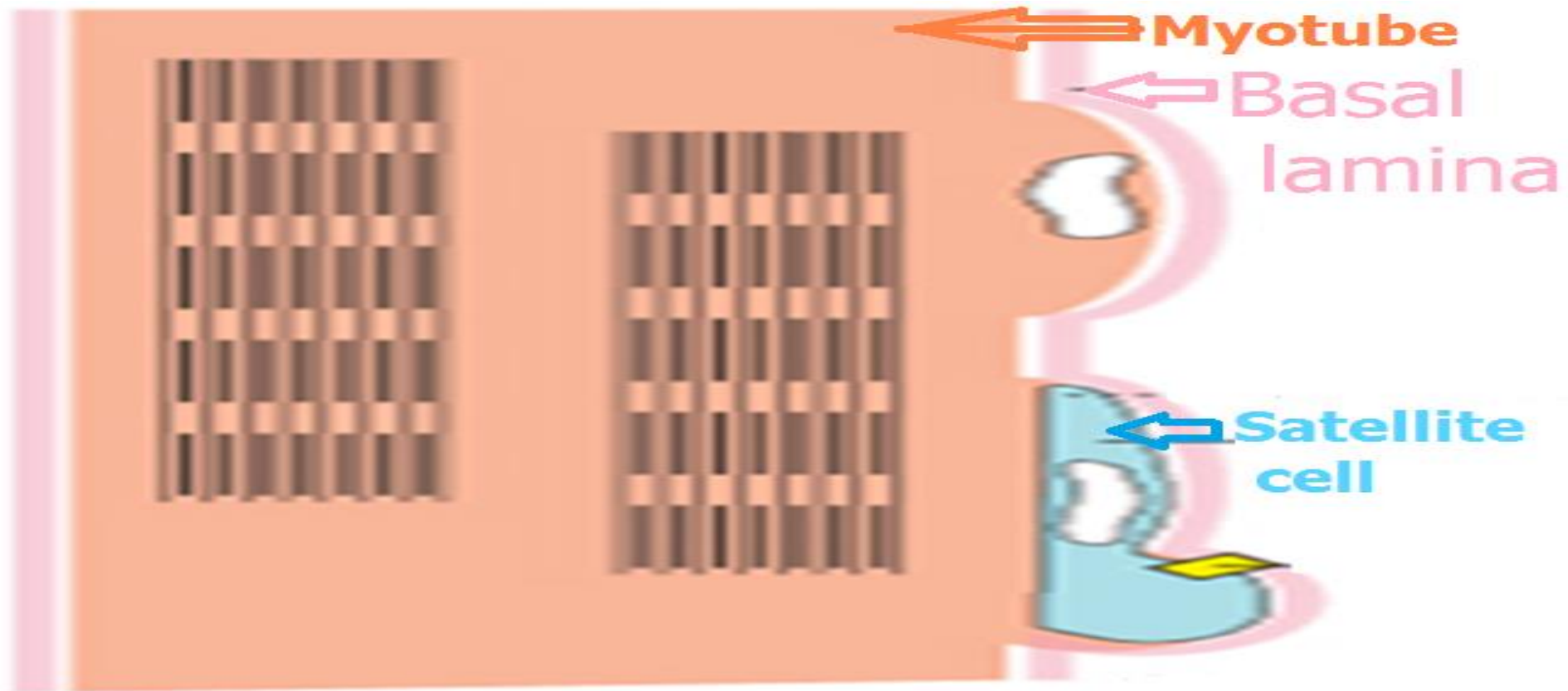
Satellite cells (myoblast like cells) that probably are left over from embryonic development)

Satellite cells



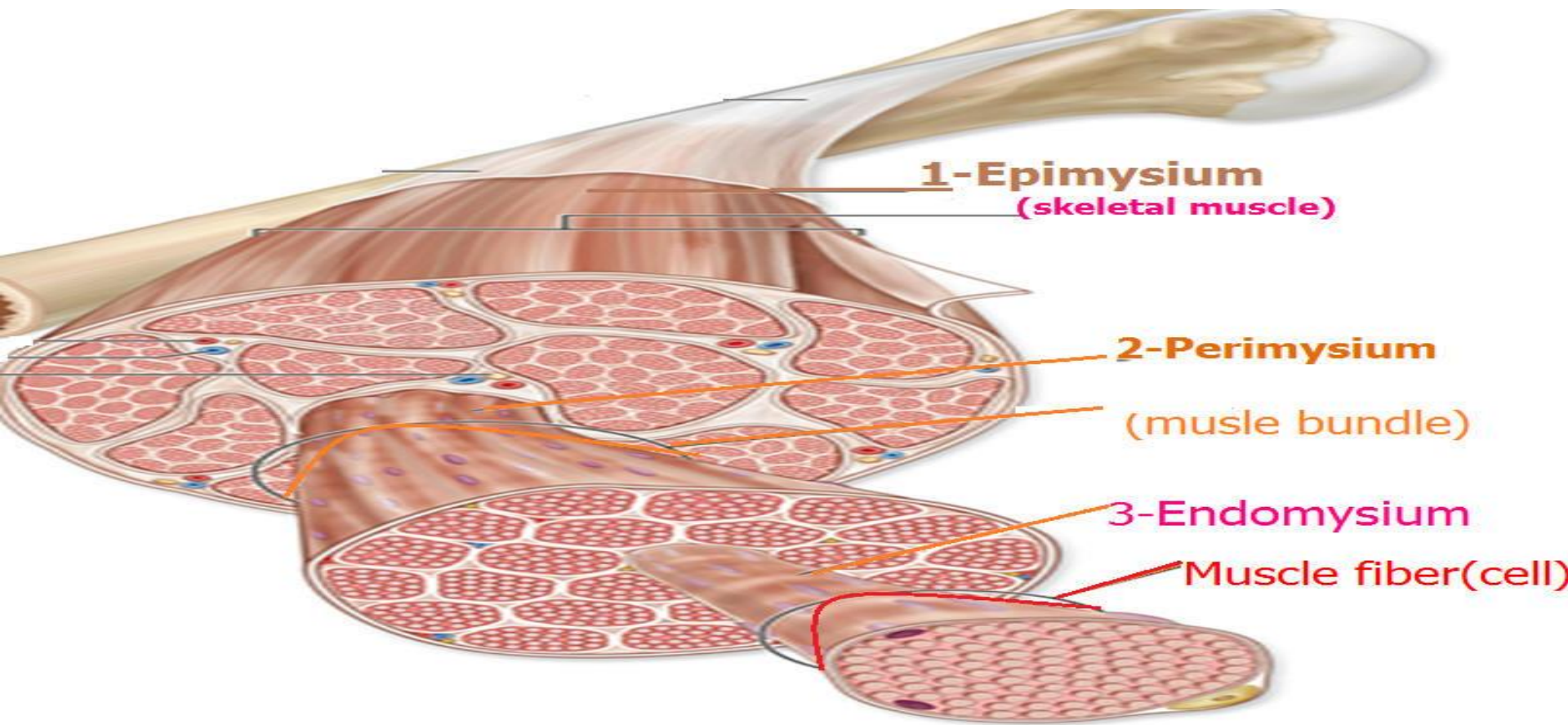
Satellite cell lie within the external lamina (basal lamina) of **skeletal muscle cells**.

Satellite cells



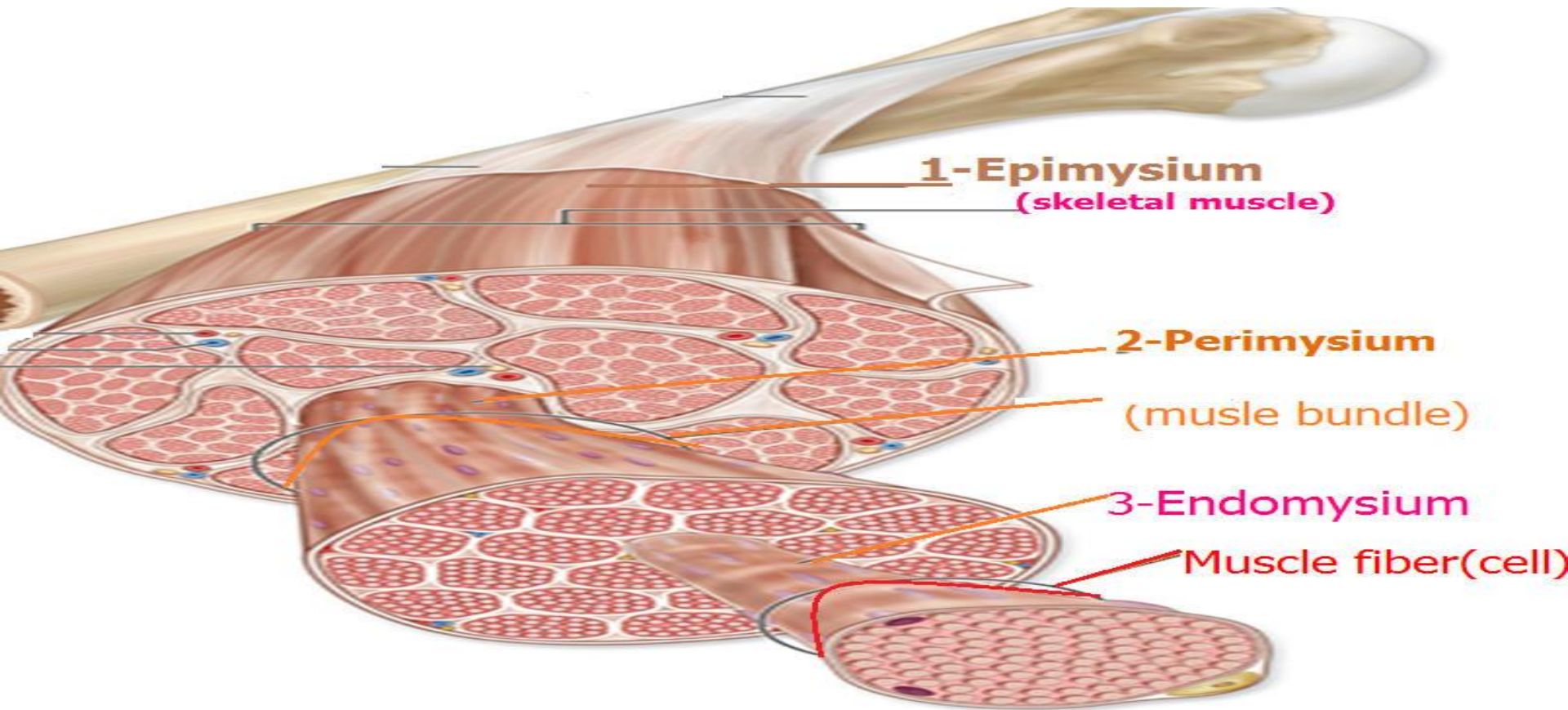
Satellite cells regenerative cells differentiate, fuse with one another, & form **skeletal muscle** cells when the need arises.

C.T.of skeletal muscle



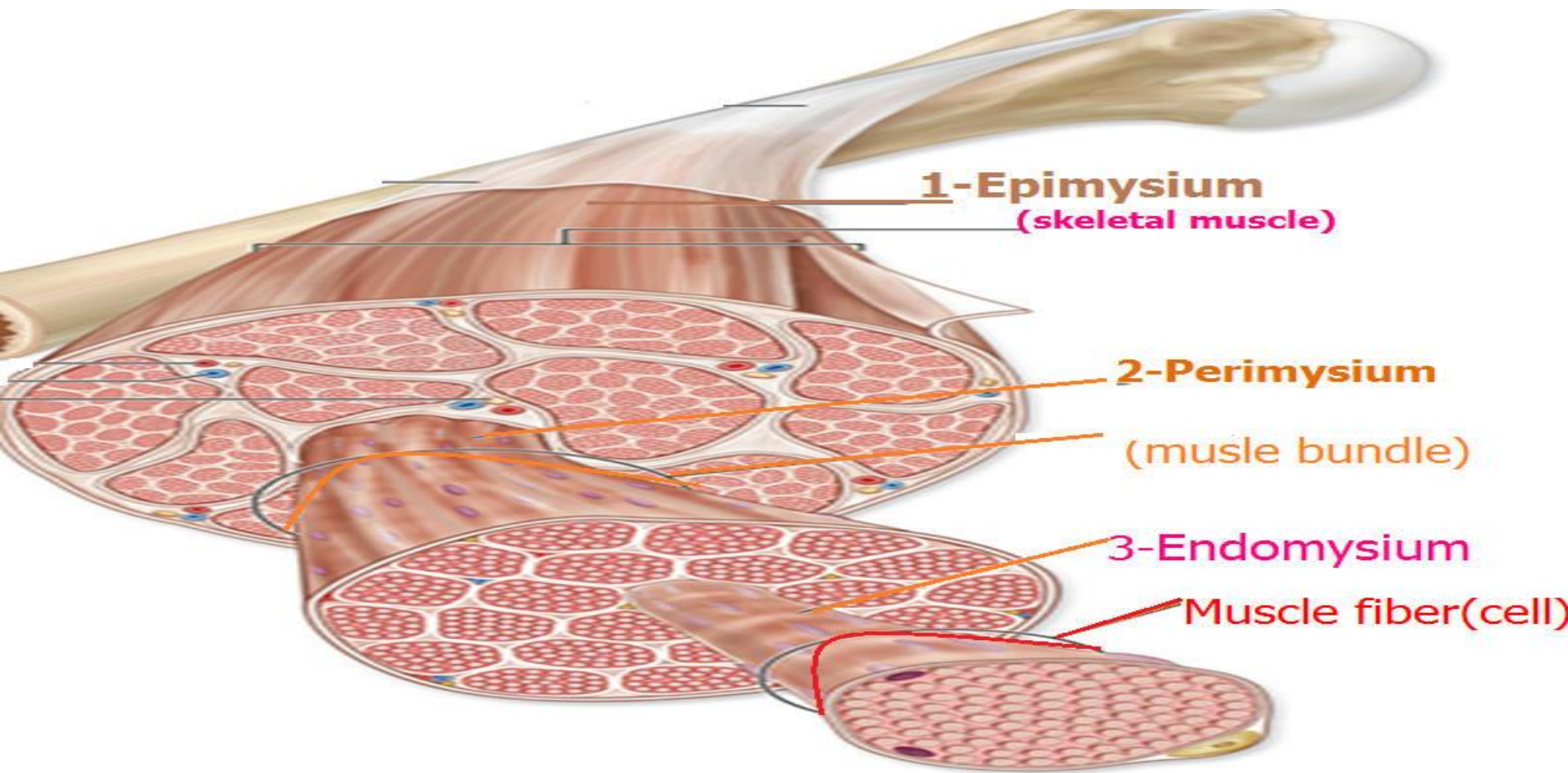
1-Epimysium (Epi=upper, + mys= muscle), an external sheath of dense C.T. surrounding the entire muscle .

C.T.of skeletal muscle



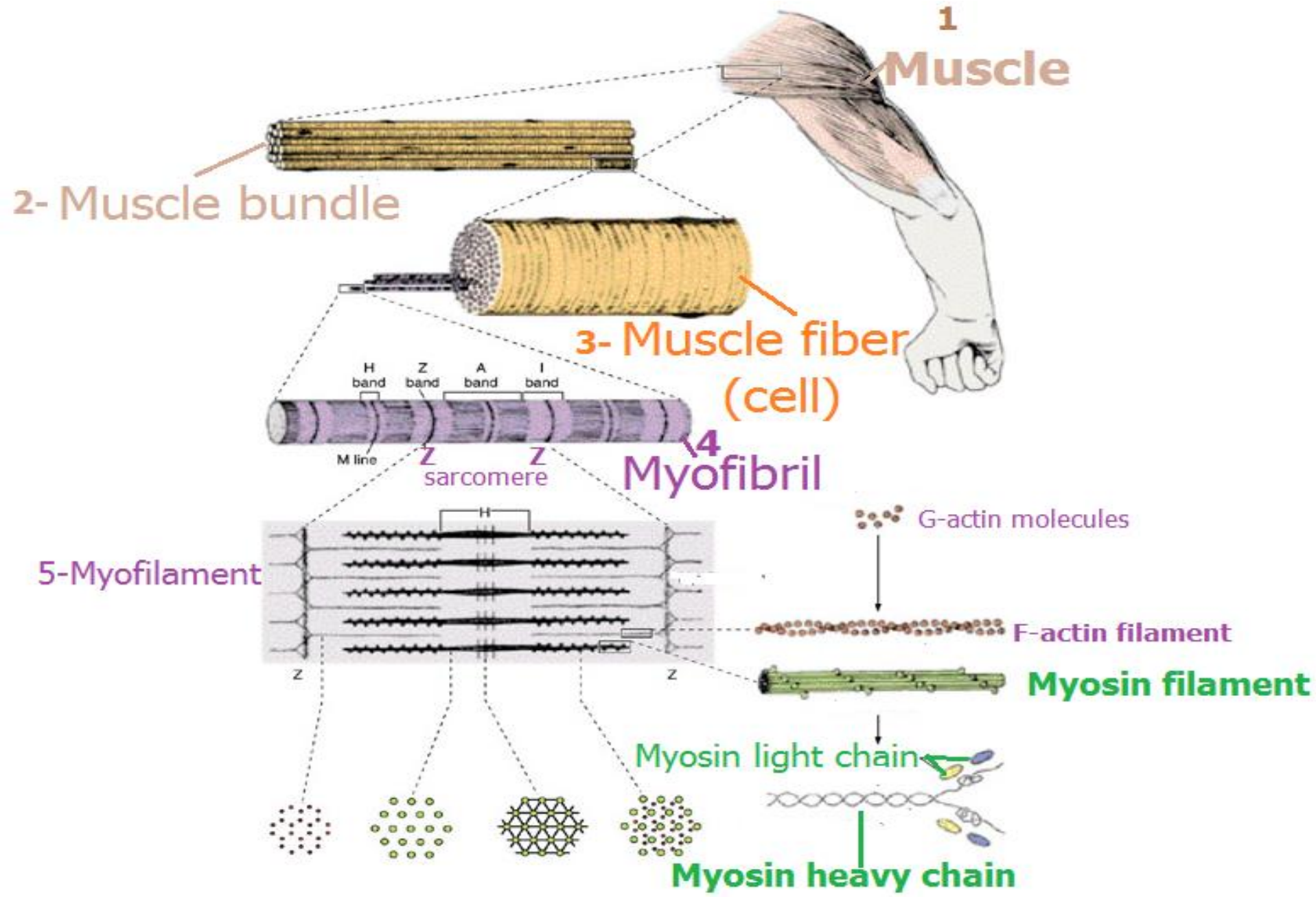
2-Perimysium (Gr. Peri= around). The C.T. around each bundle of muscle fibers

C.T.of skeletal muscle



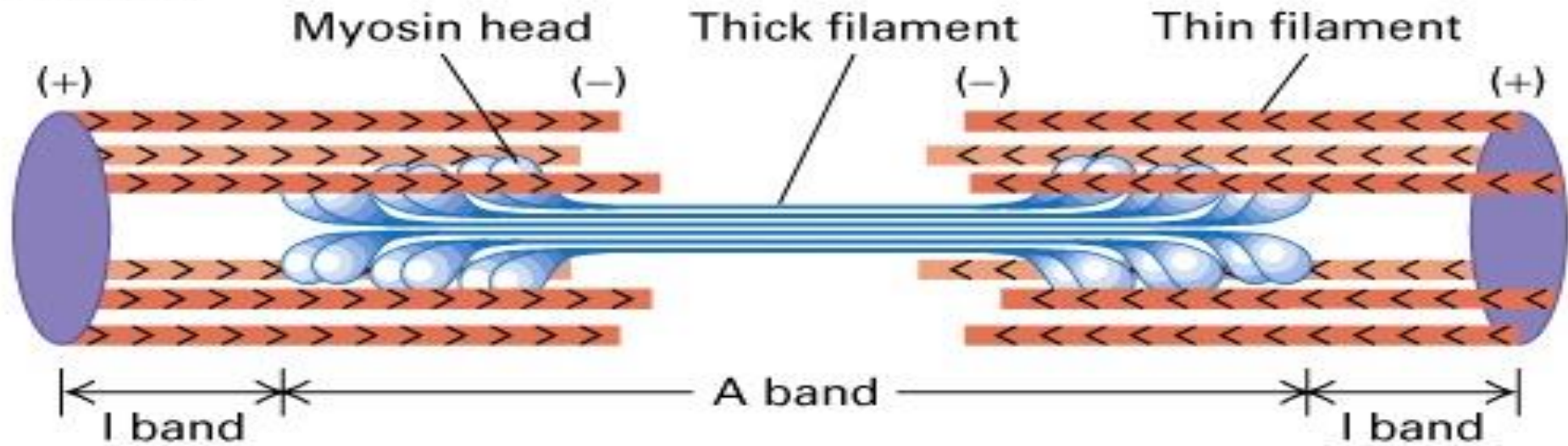
3-Endomysium (Gr. Endon=**within**), composed mainly of a **basal lamina** and **reticular fibers** surrounding each muscle fiber

Organization of skeletal muscle



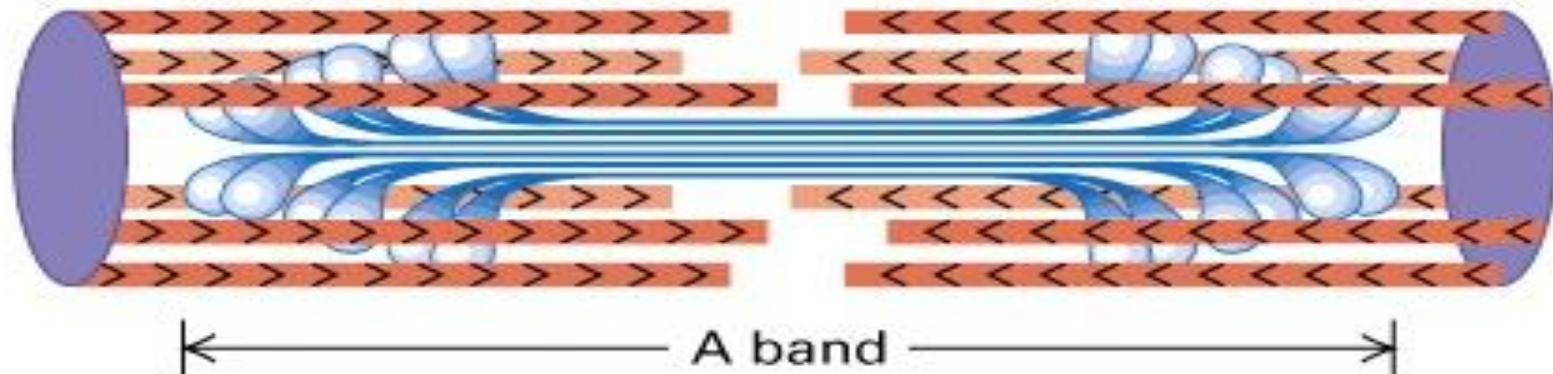
Contraction of striated muscles

Relaxed

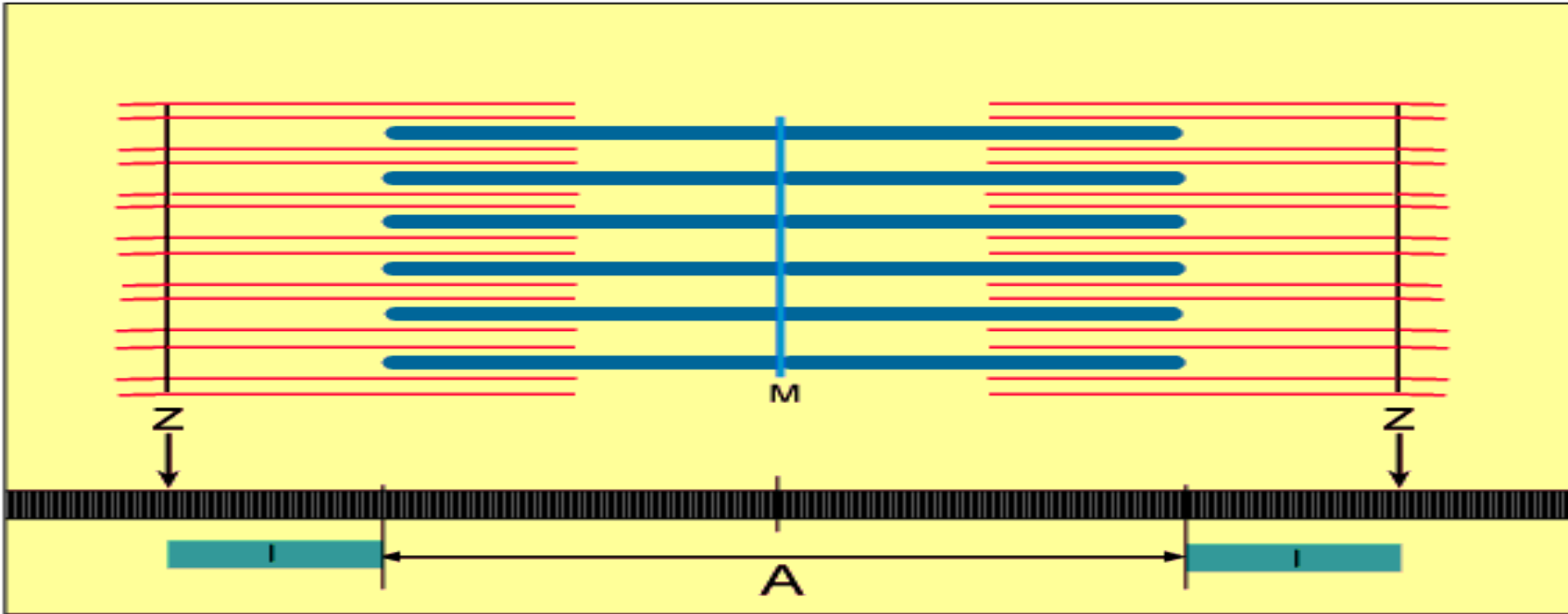


+ATP, Ca^{2+}

Contracted

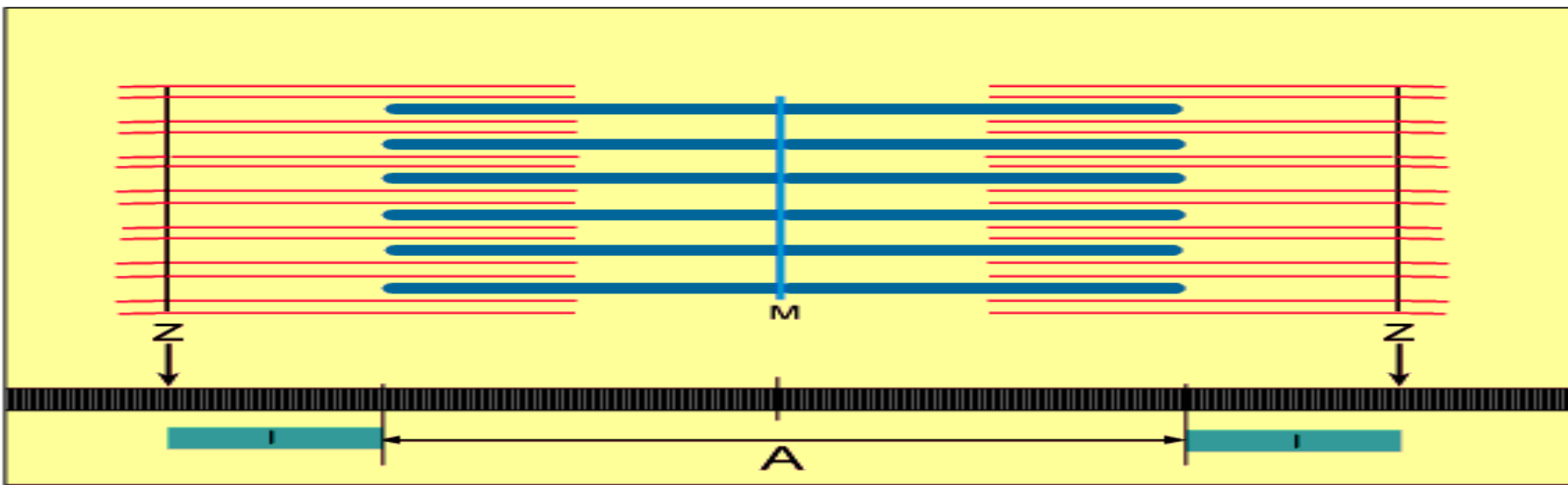


Sarcomere



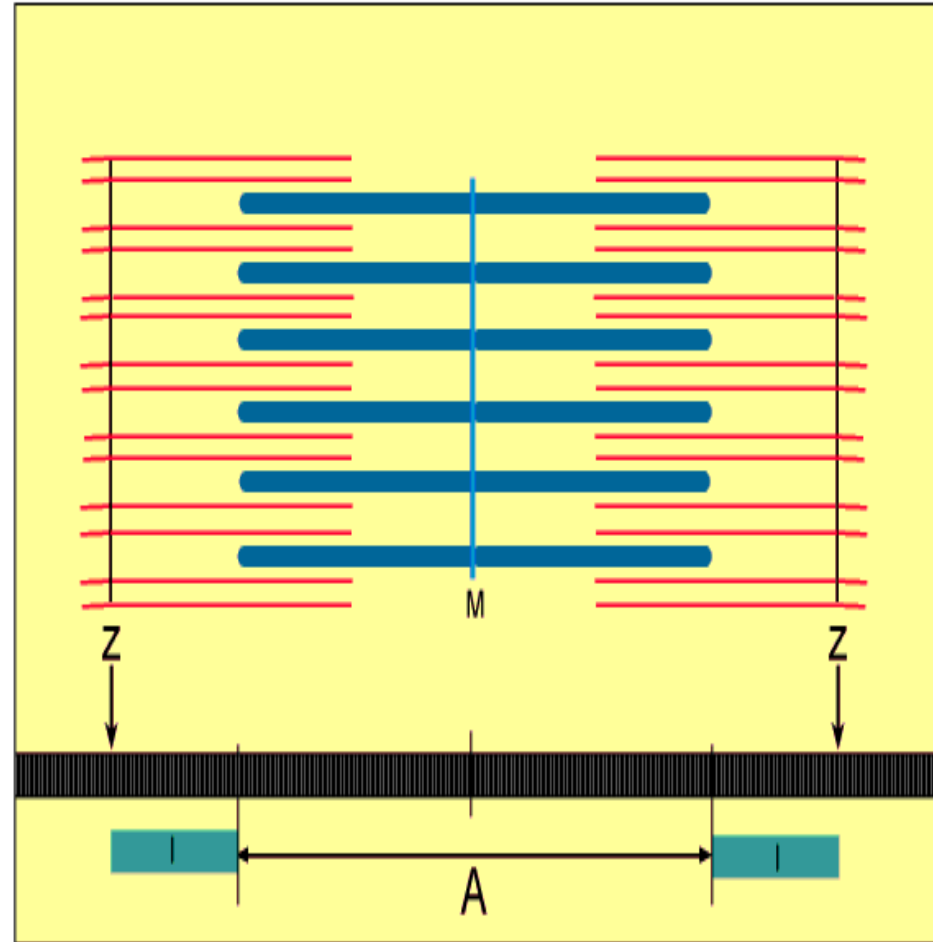
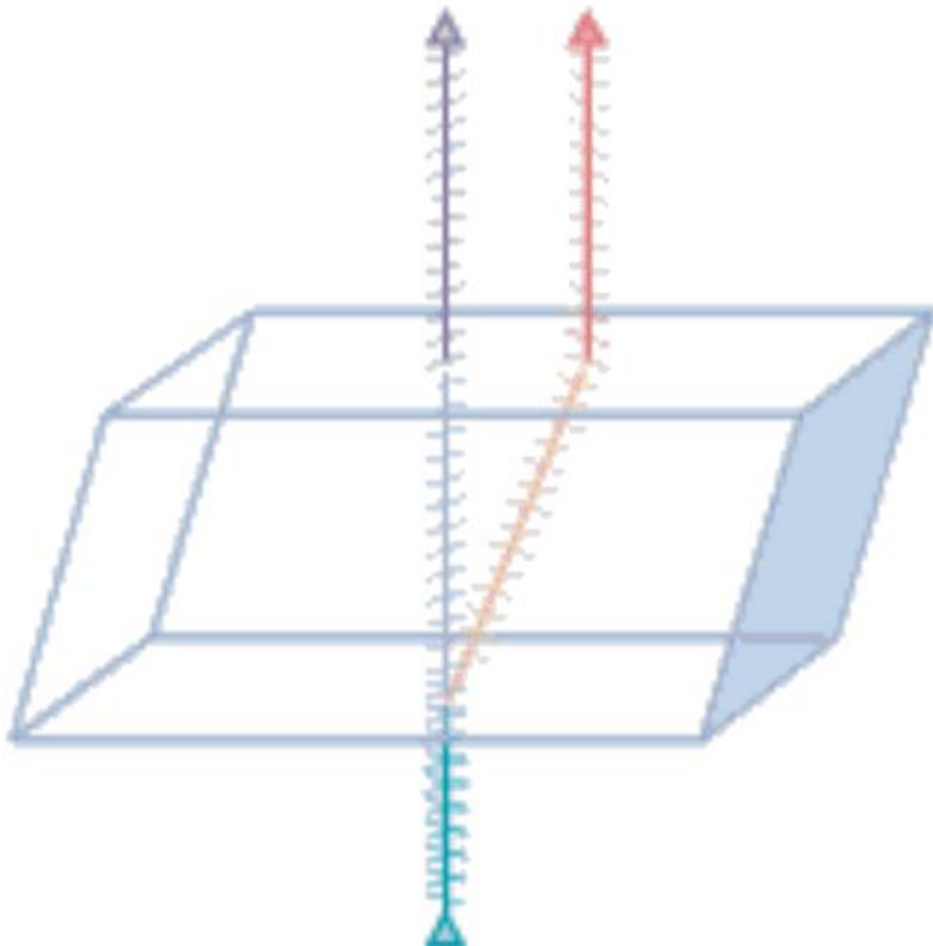
The **sarcomere** is the smallest repetitive subunit of the contractile apparatus, it extends from Z line to Z line & is about $2.5\mu\text{m}$ long in resting muscle.

Sarcomere



- The darker bands are called **A** bands (anisotropic, are birefringent in polarized light).
- The lighter bands are called **I** bands (isotropic, do not alter polarized light).
- Each I band is bisected by a dark transverse line, the Z line.

Anisotropic

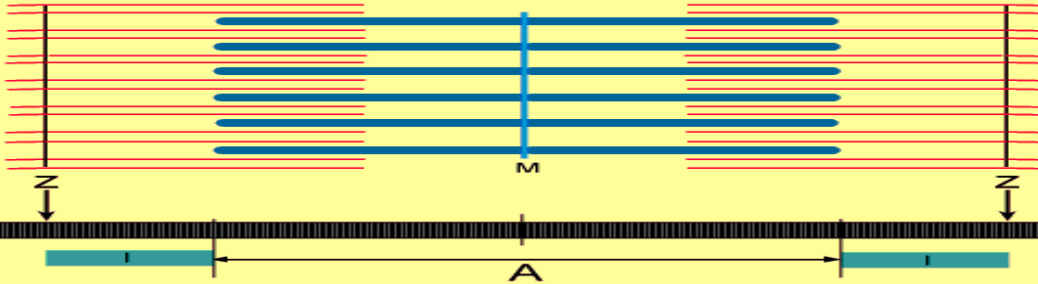


Anisotropic, are birefringent in polarized light.

Bands

A band

I band



Color

Dark

Light

Filaments

Thin

+

thick

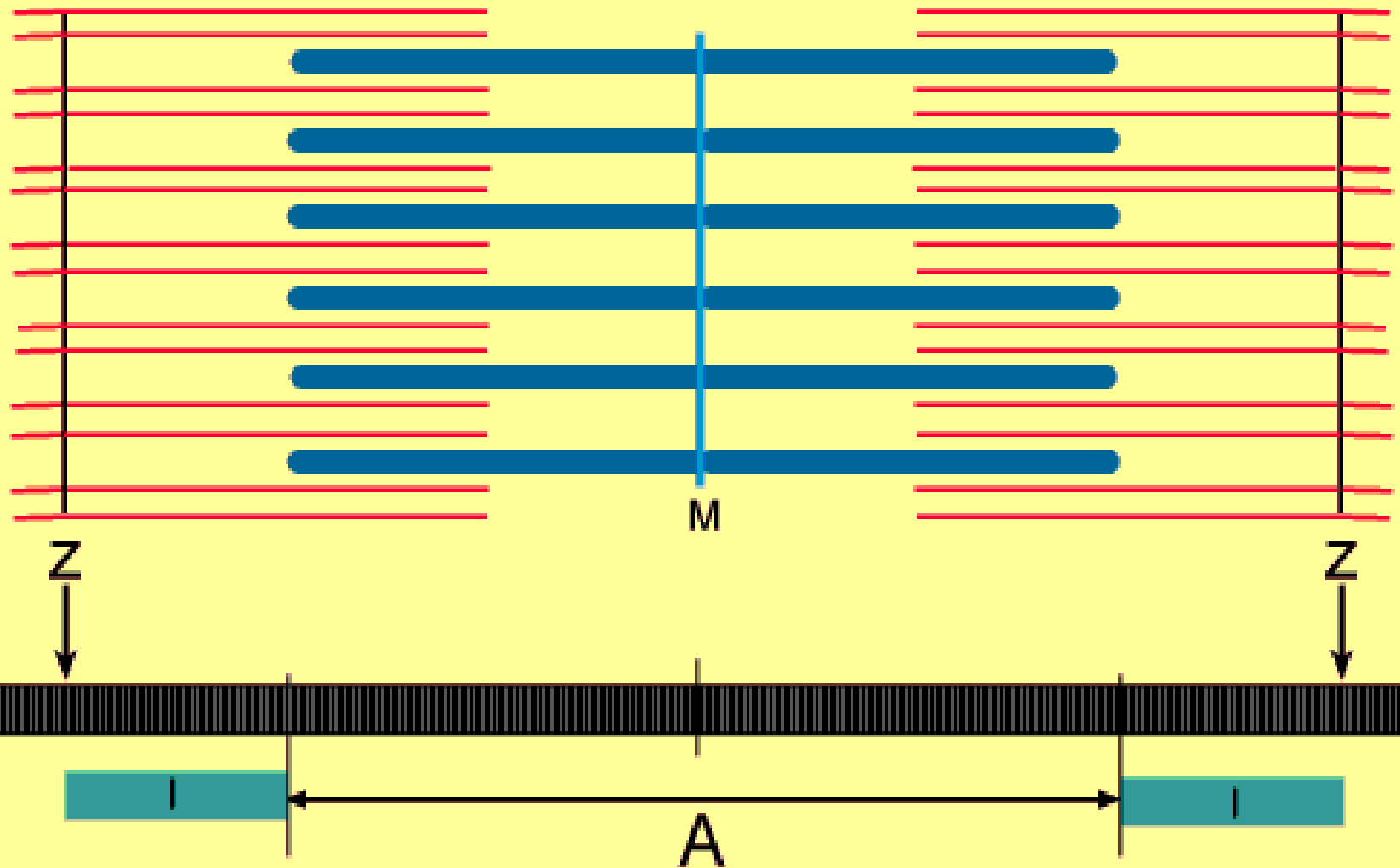
Thin

Divided by:

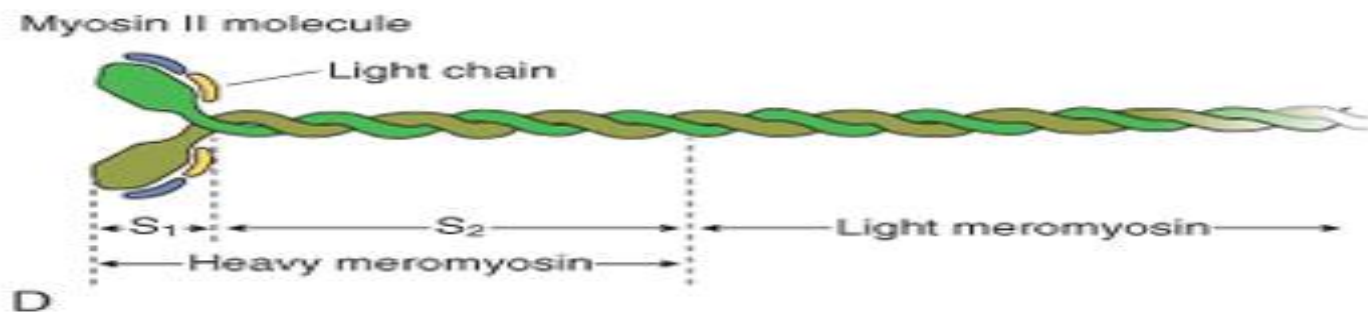
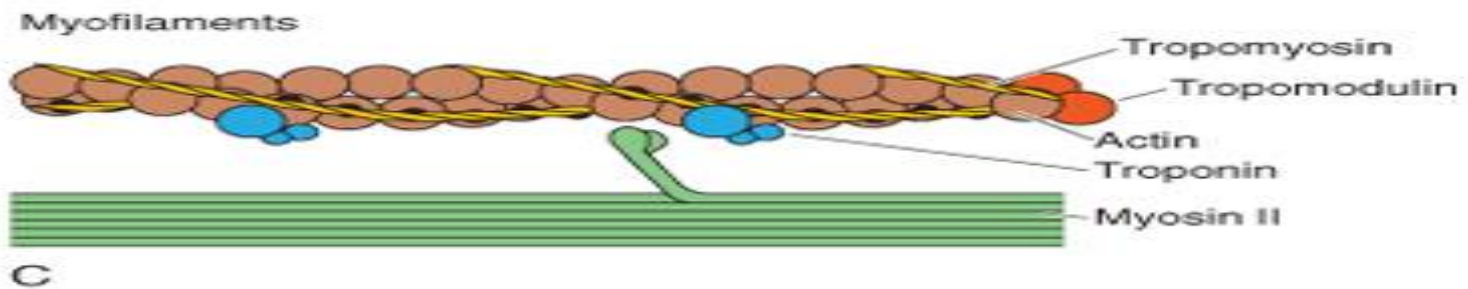
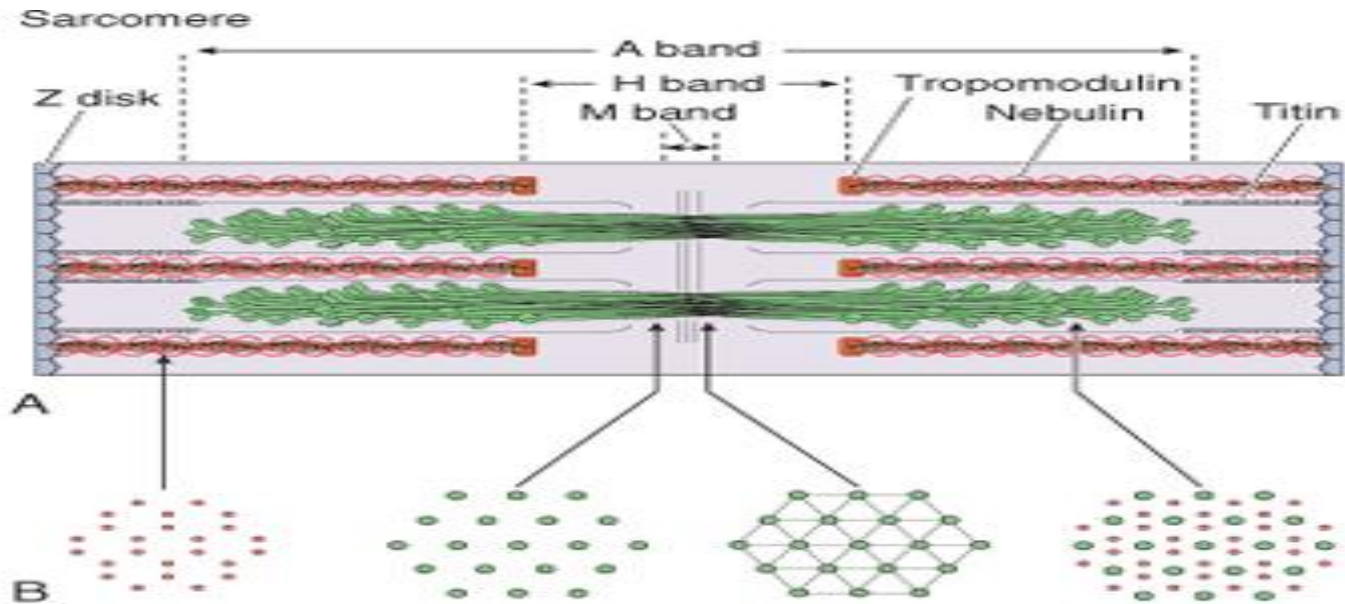
M line.

Z line.

Sarcomere

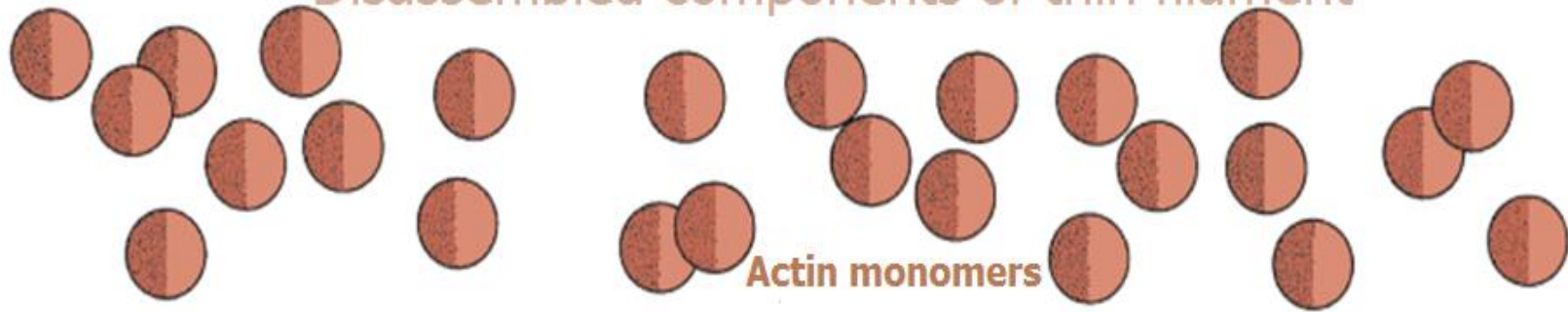


Sarcomere

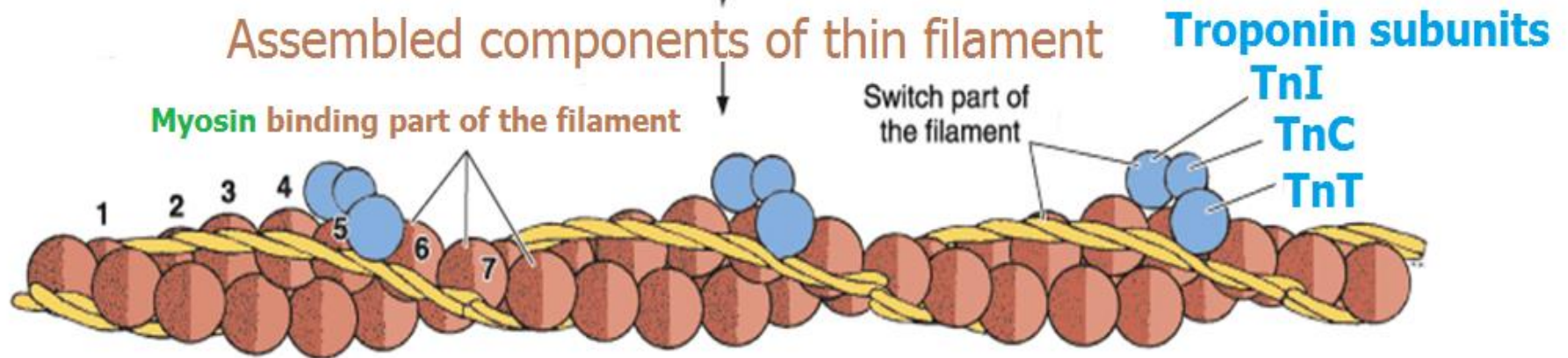


Thin filament

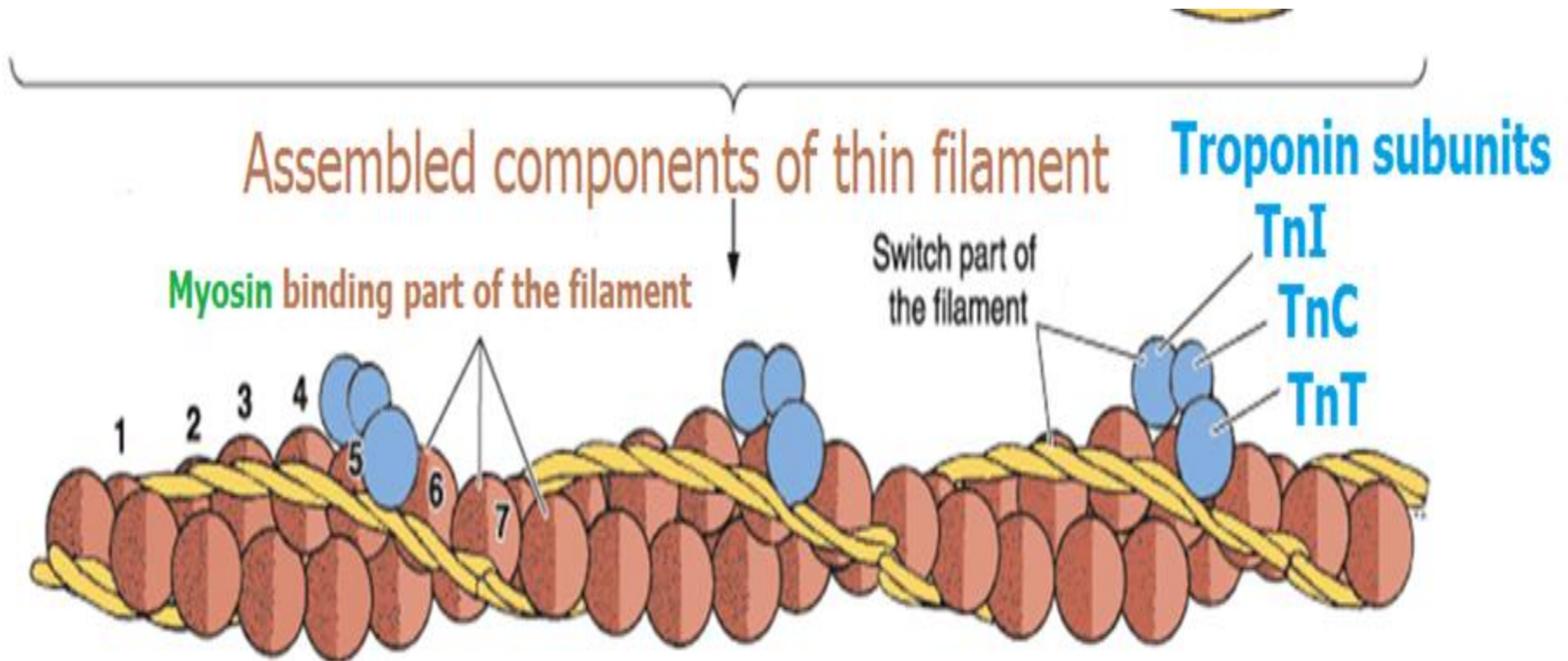
Disassembled components of thin filament



Assembled components of thin filament

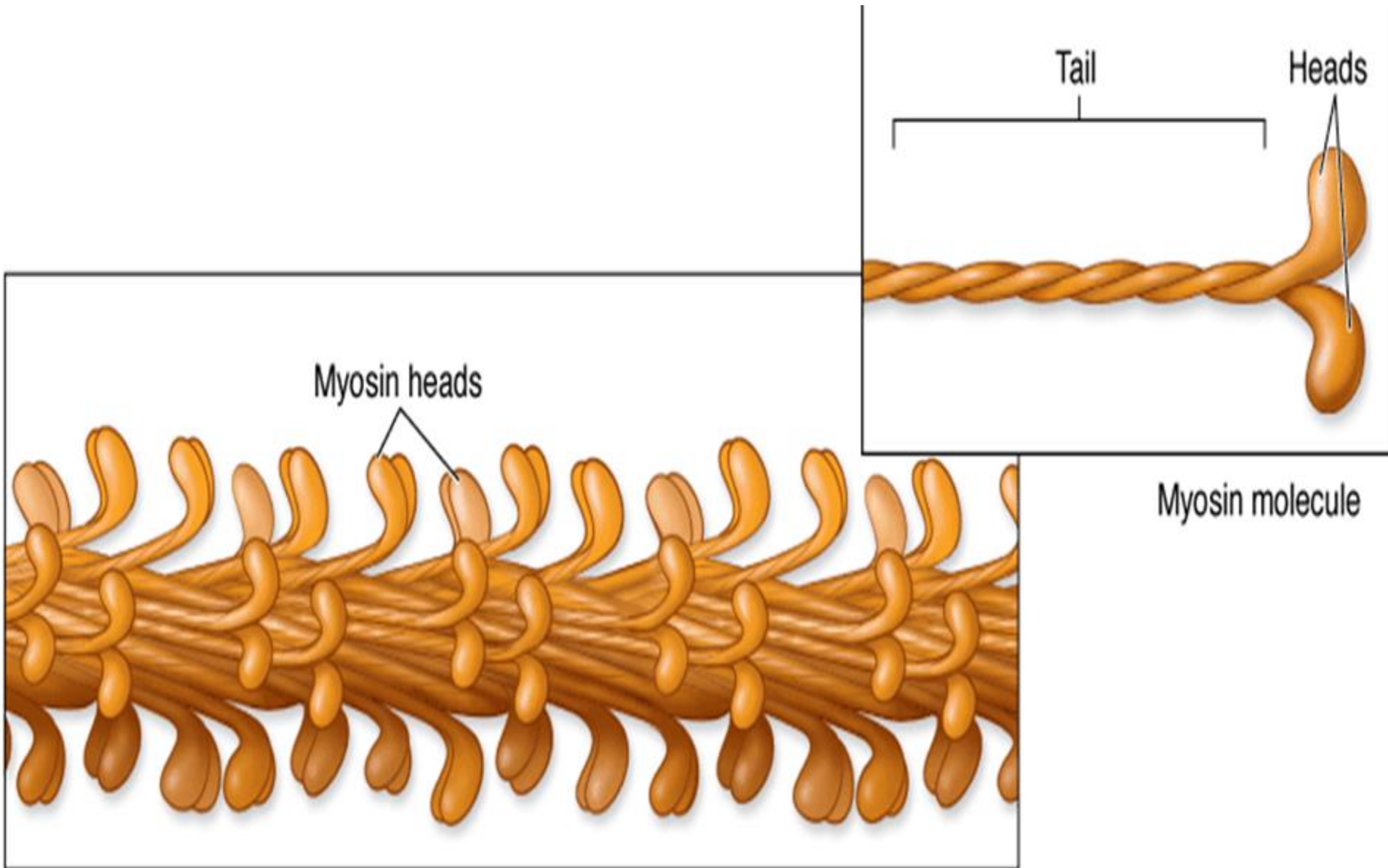


Troponin complex of 3 subunits:

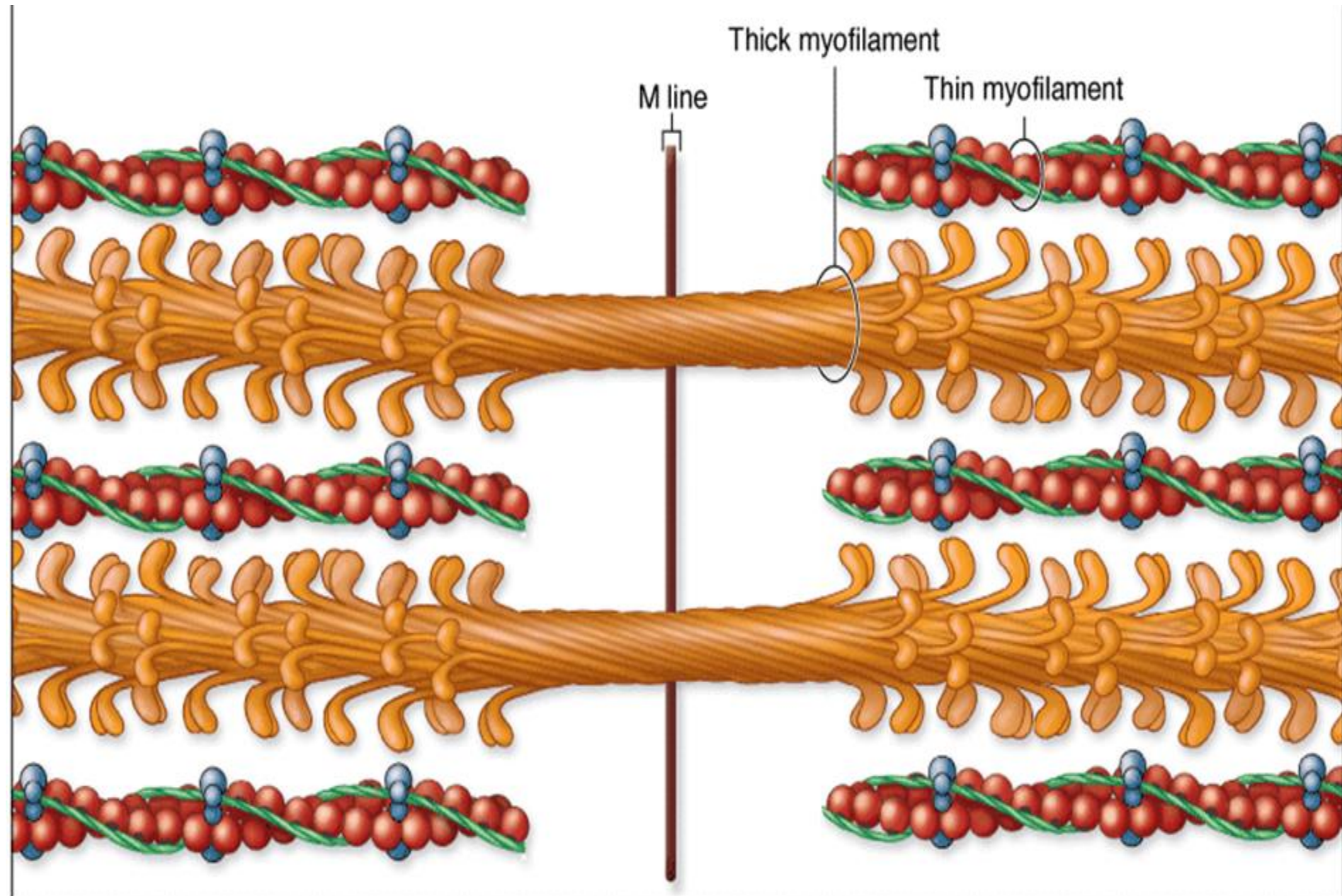


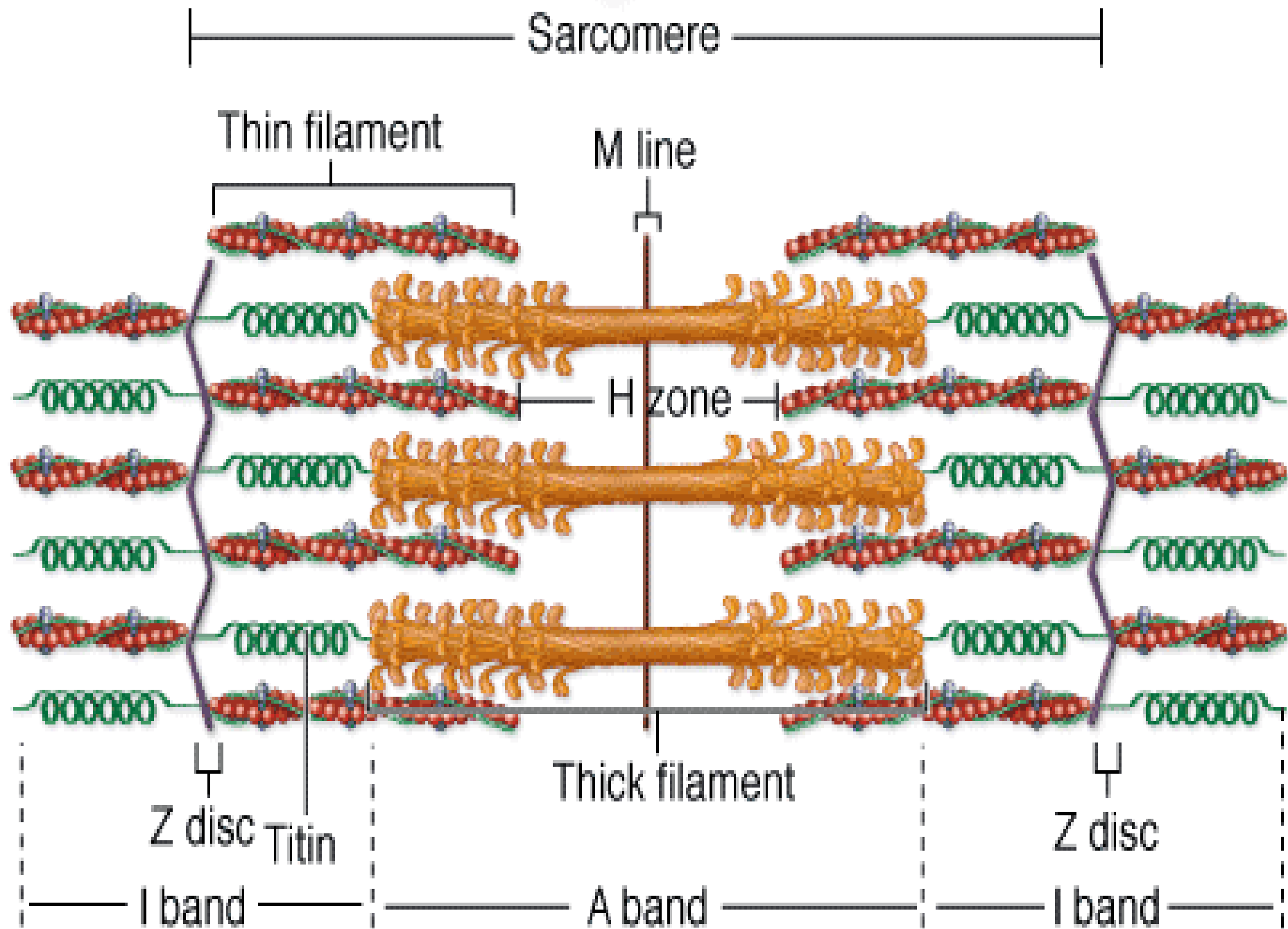
TnT, which strongly attaches to **tropomyosin**;
TnC, which binds **calcium** ions;
TnI, which **inhibits the actin myosin interaction**.

Thick filament

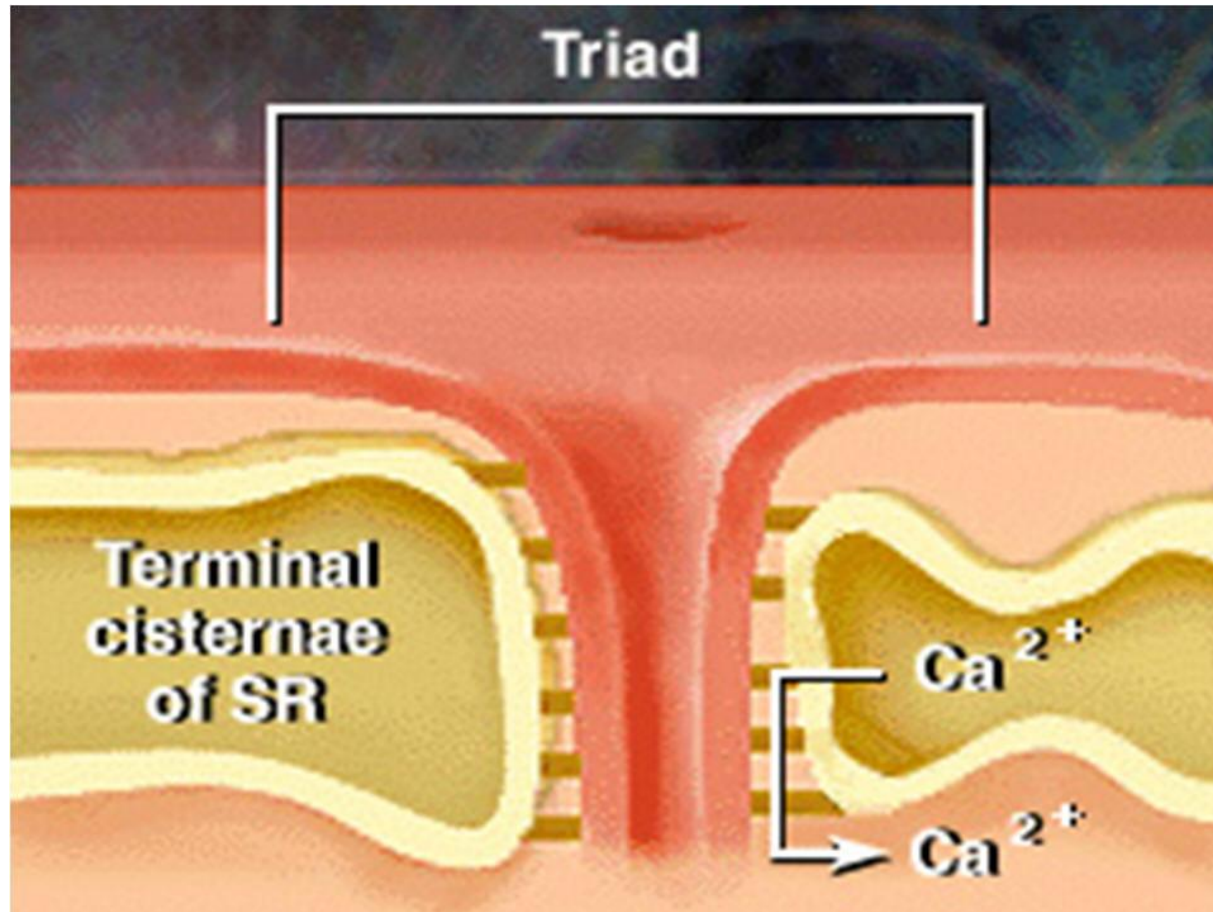


Comparison of thick & thin filaments

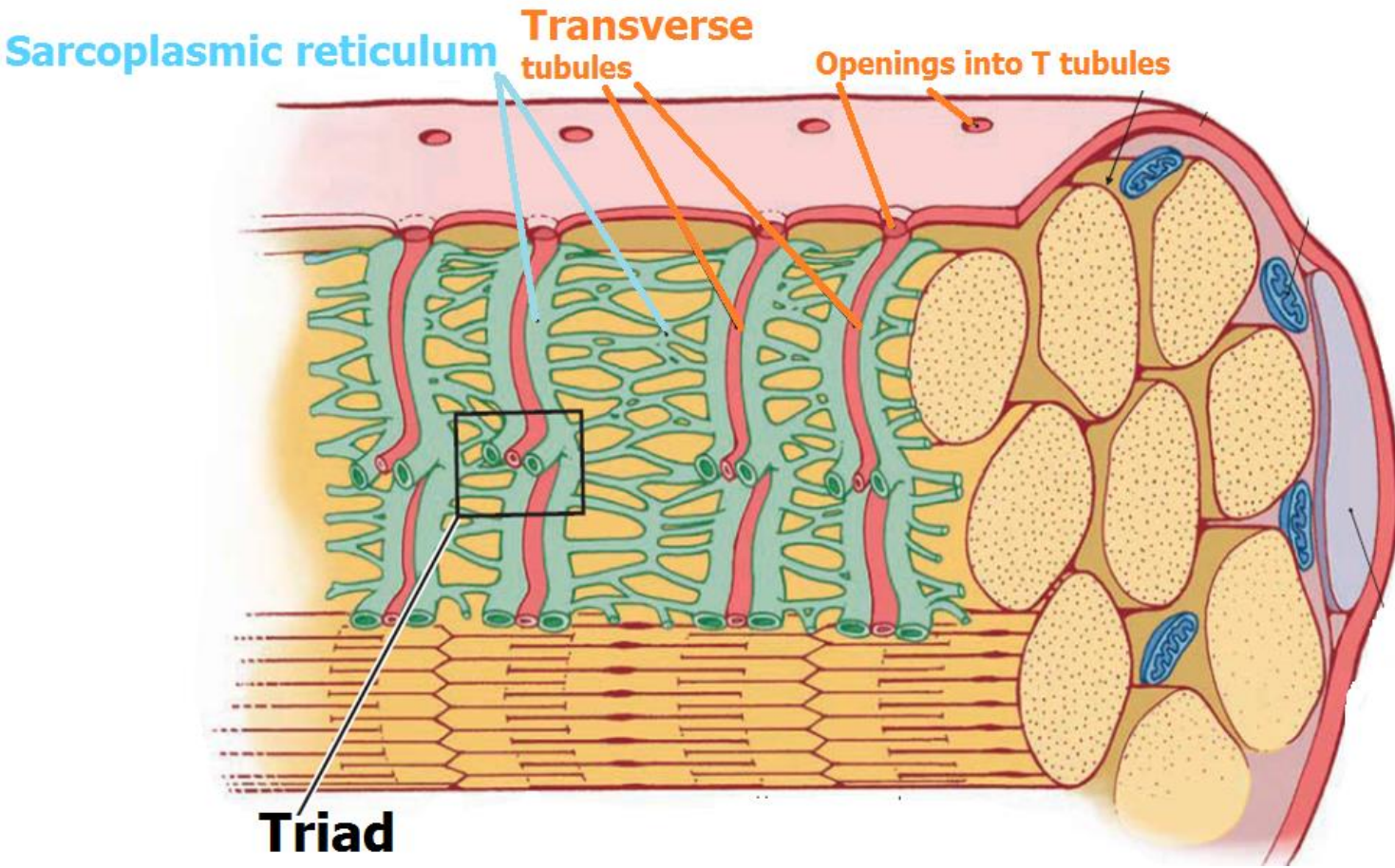




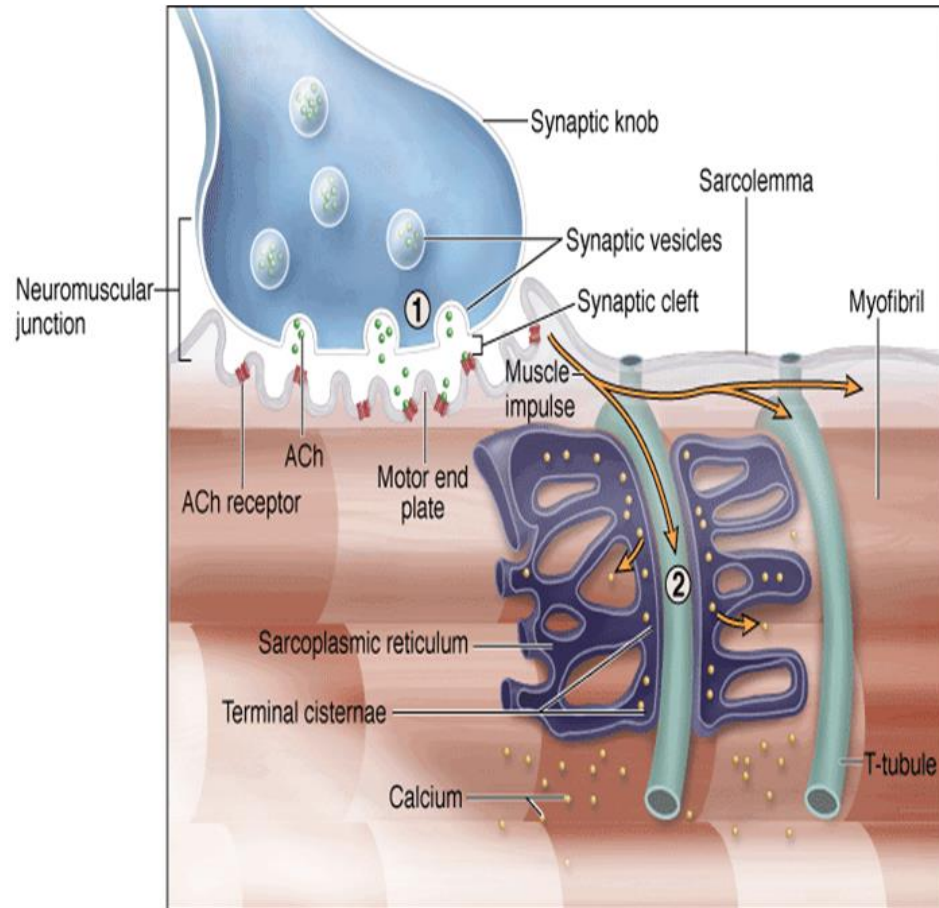
Organization of triads of skeletal muscle



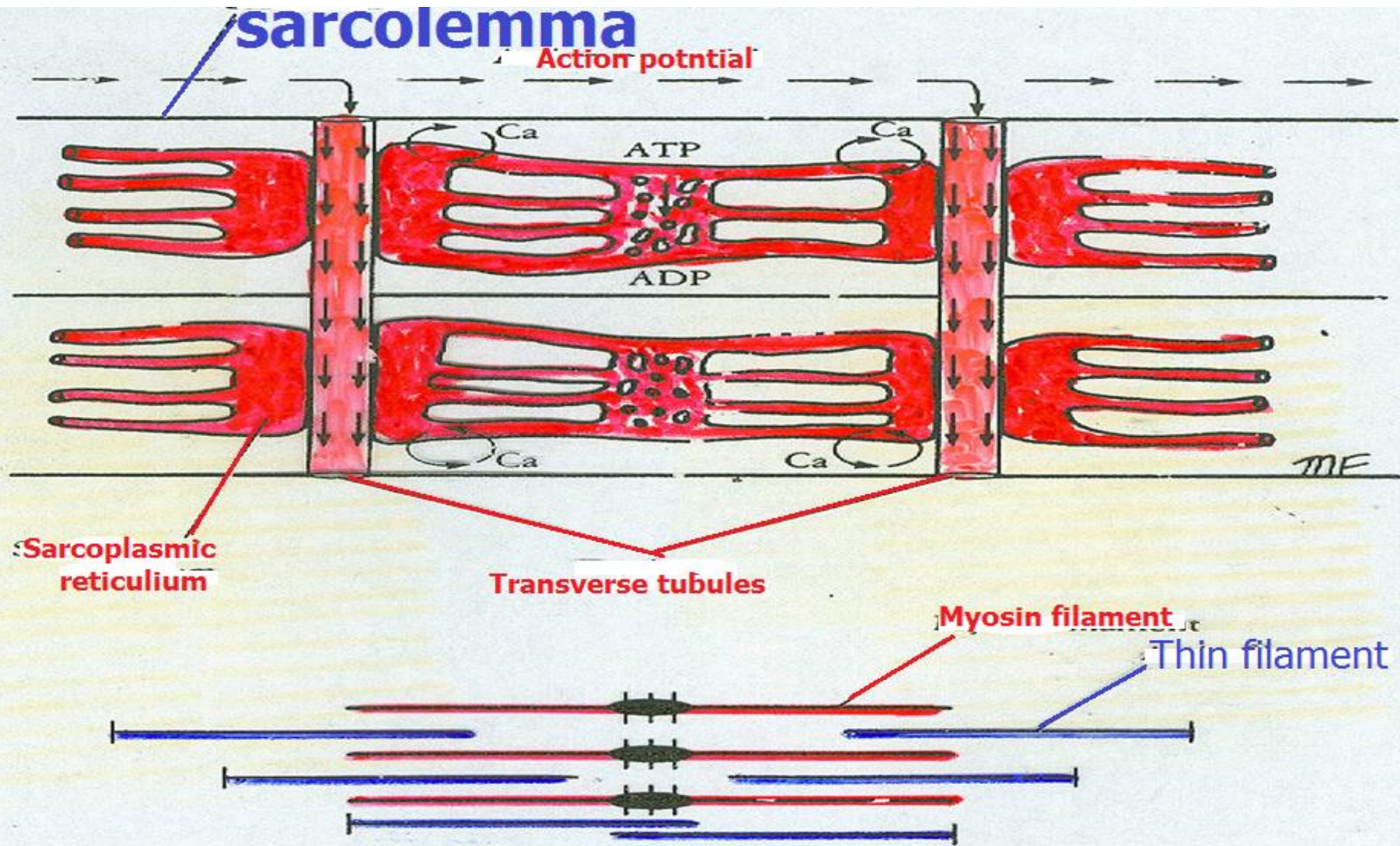
Organization of triads & sarcomeres of skeletal muscle



Organization of triads of skeletal muscle

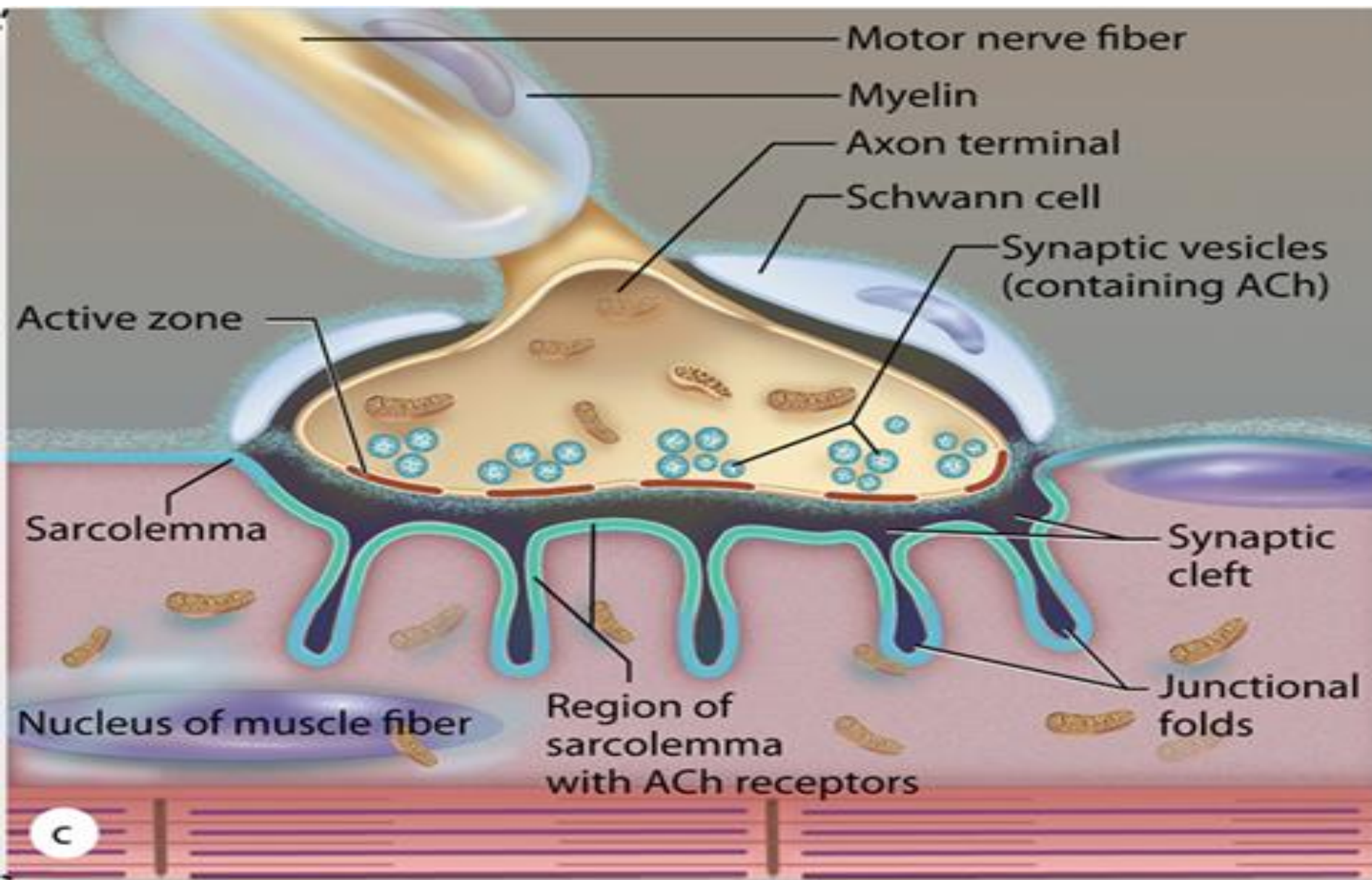


Organization of triads & sarcomeres of skeletal muscle

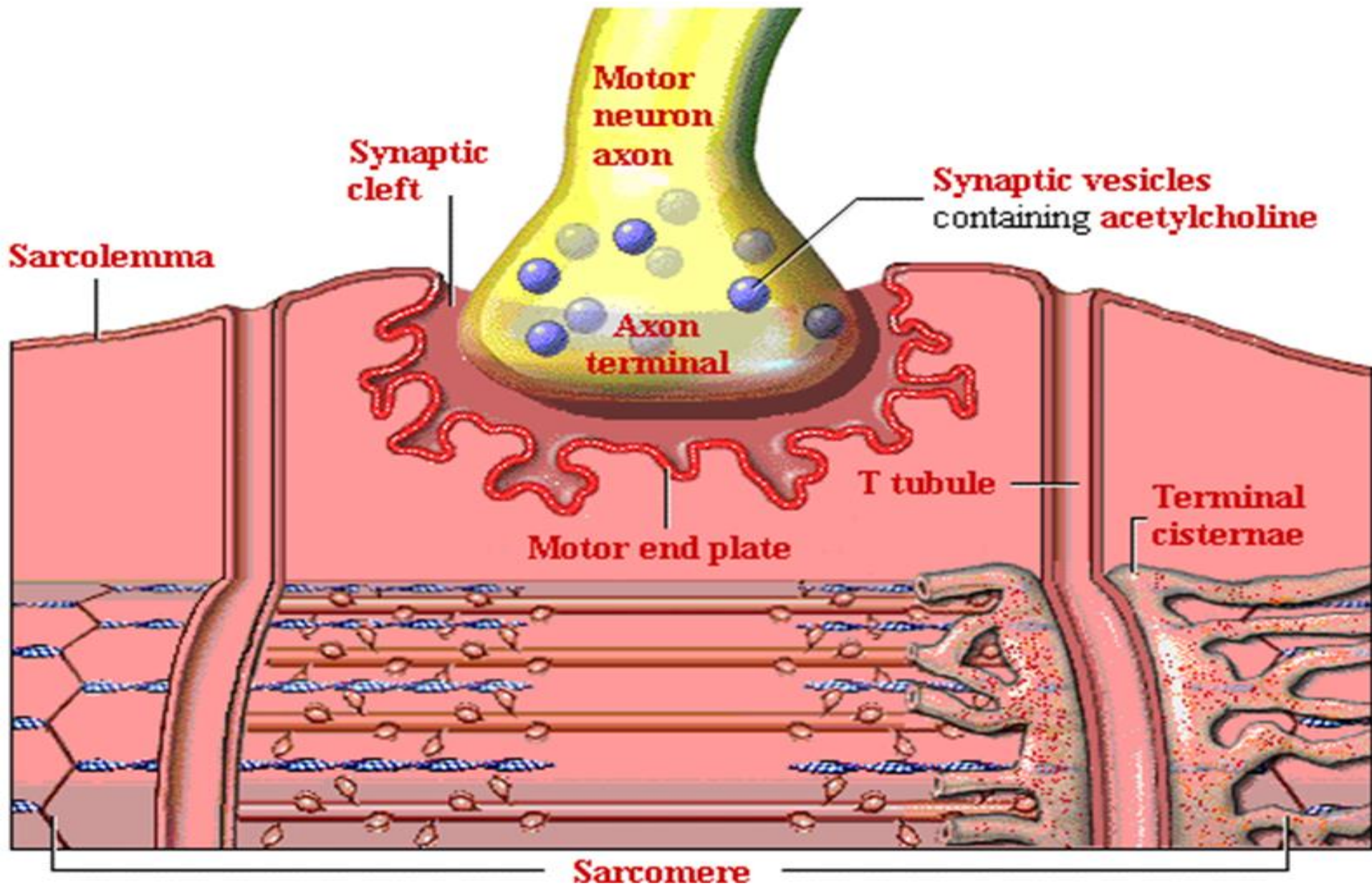


Triad present at **A-I junction** (**triad**: 1 T tubule with 2 terminal cisternae of sER).
- **2 T tubules**/sarcomere

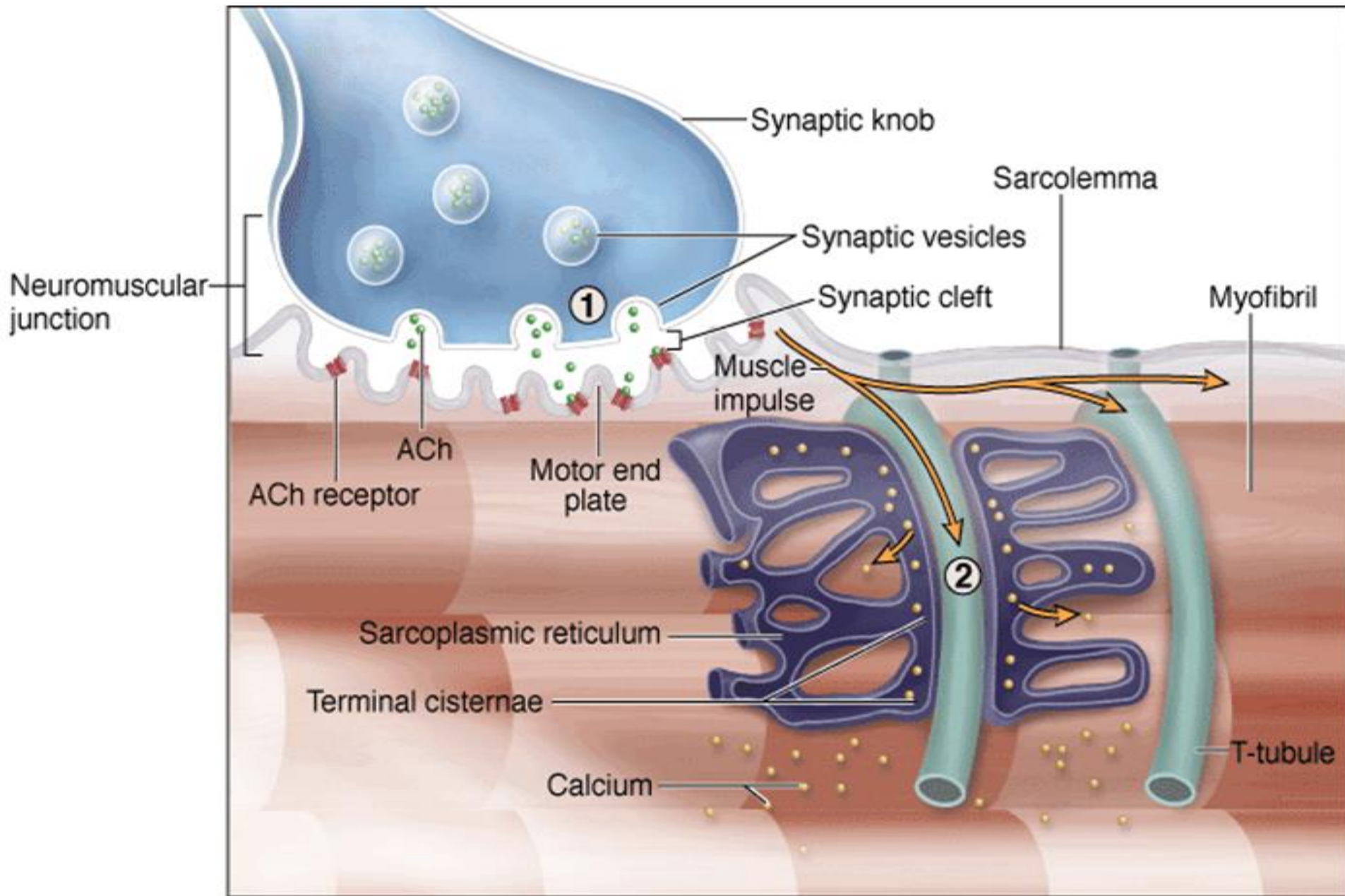
The neuromuscular junction (NMJ)



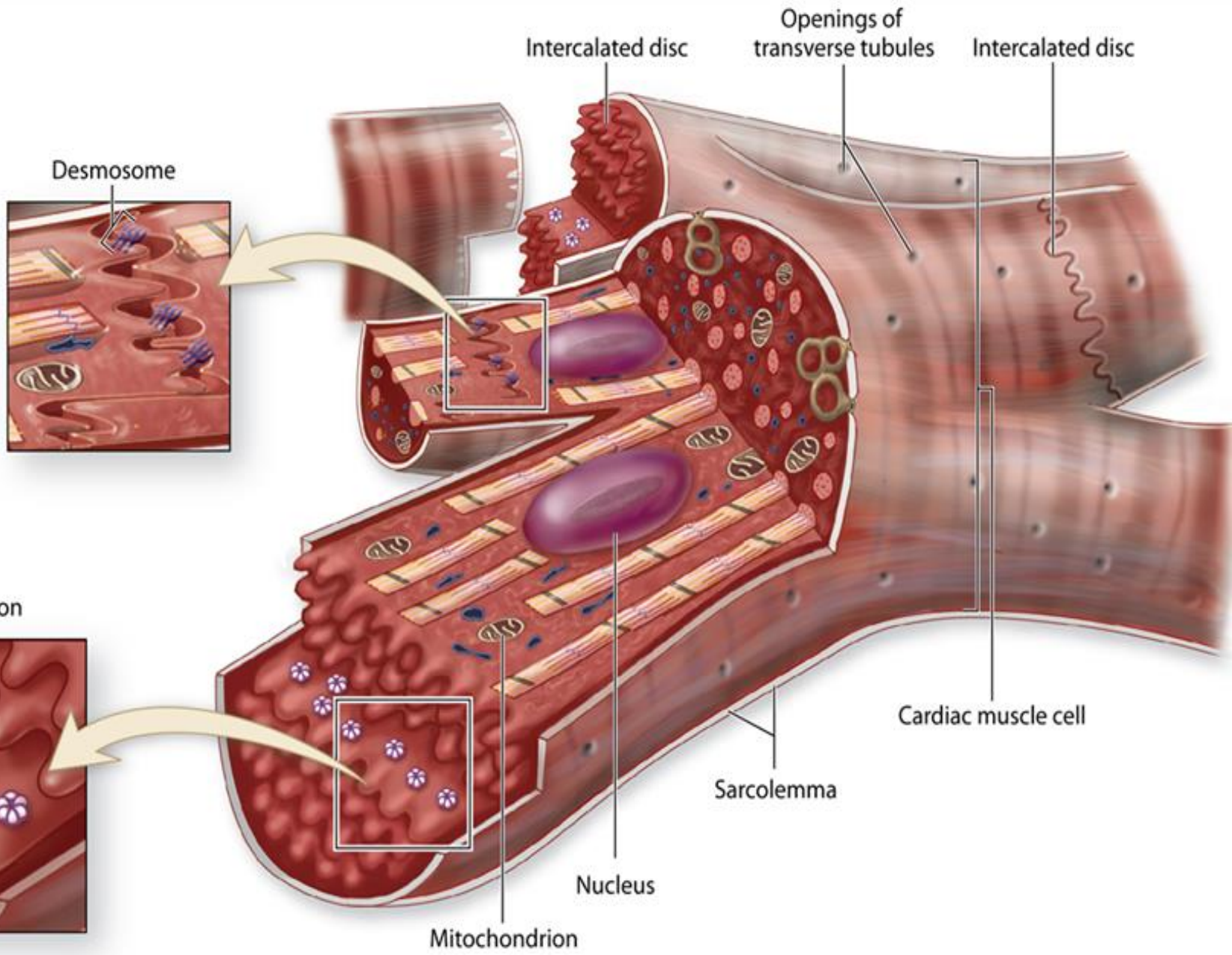
The neuromuscular junction (NMJ)



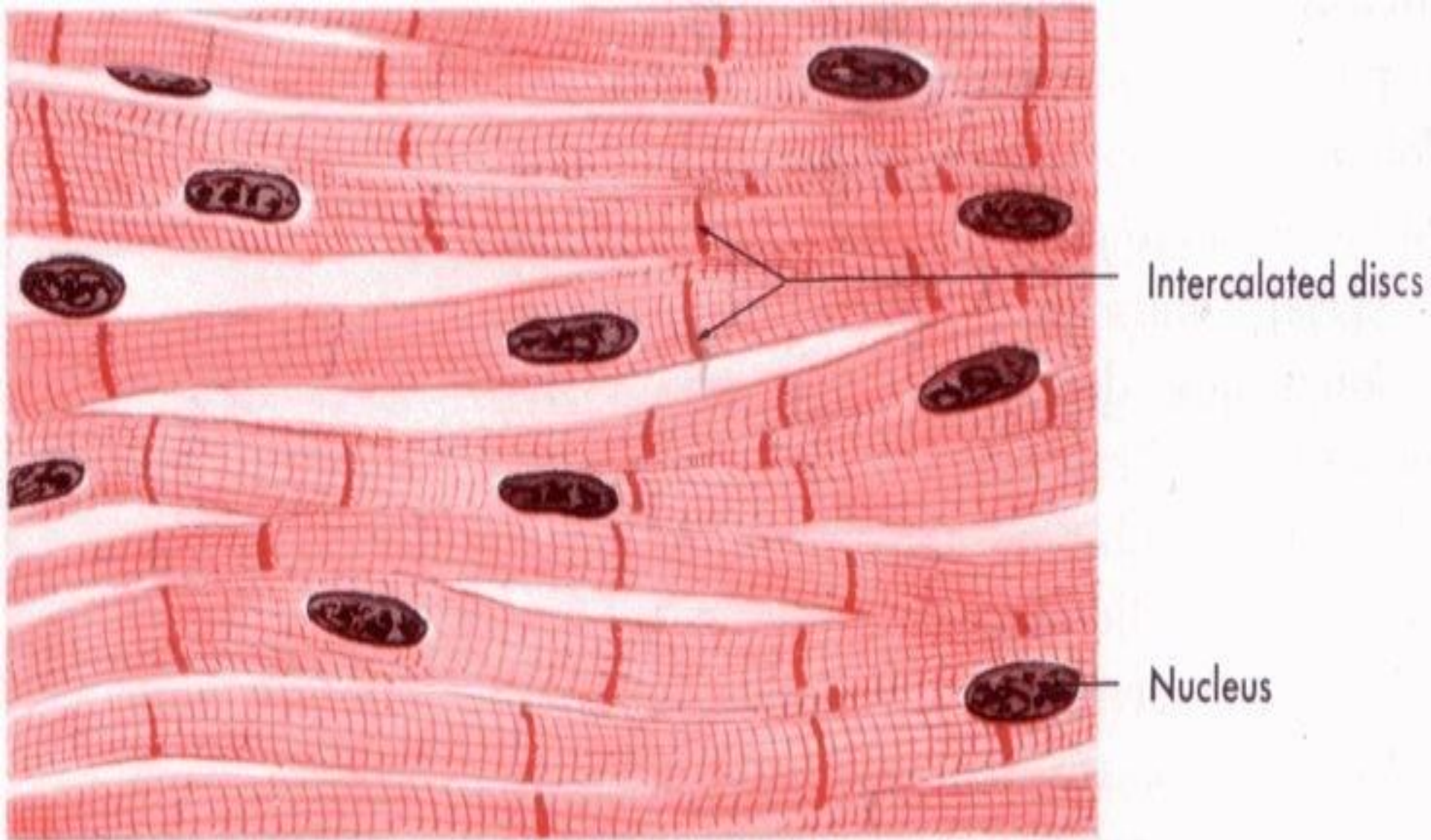
The neuromuscular junction (NMJ)



Cardiac muscle

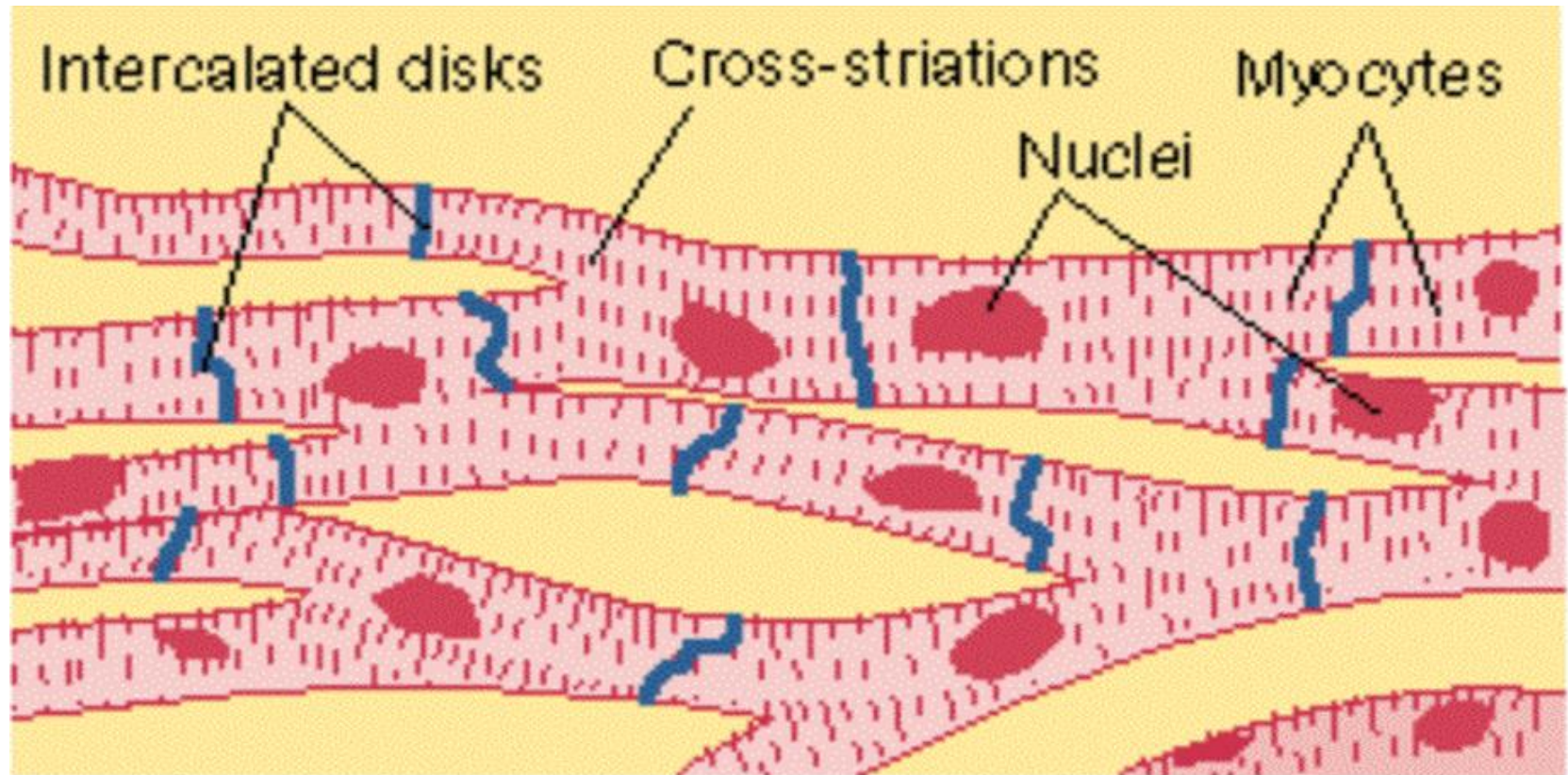


Cardiac muscle

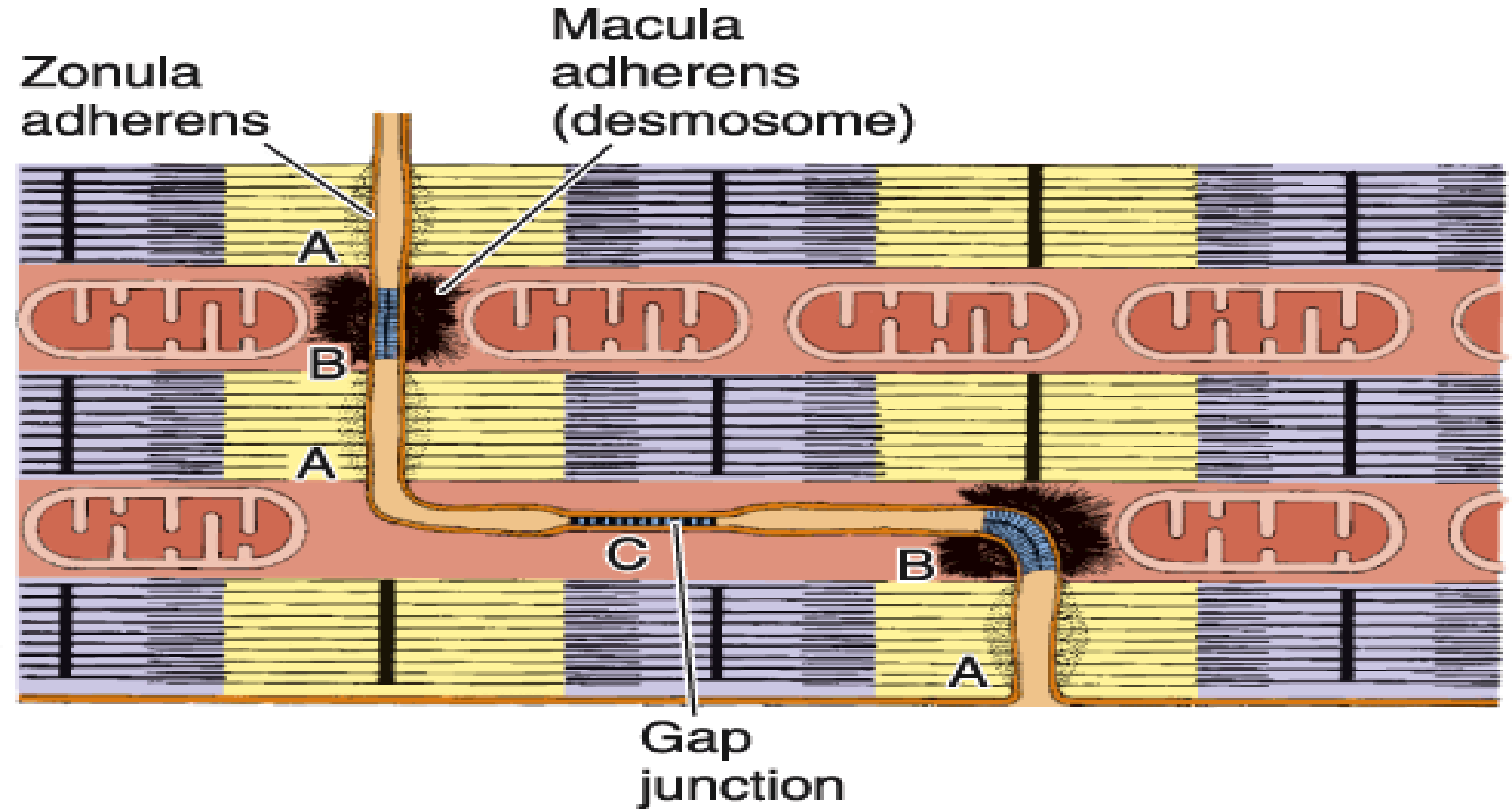


Cardiac or striated involuntary muscle tissue.

Cardiac muscles

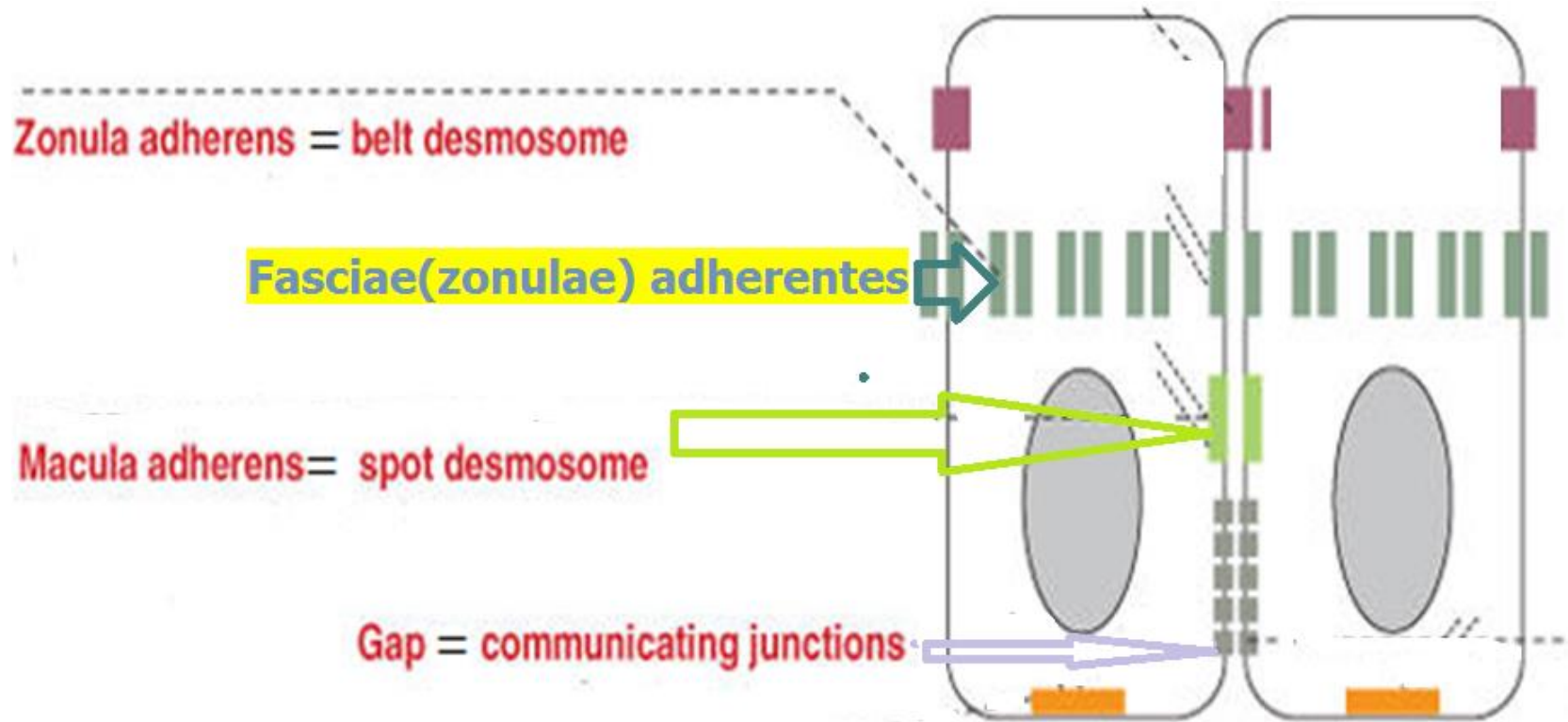


Cardiac muscle-to-muscle junctions

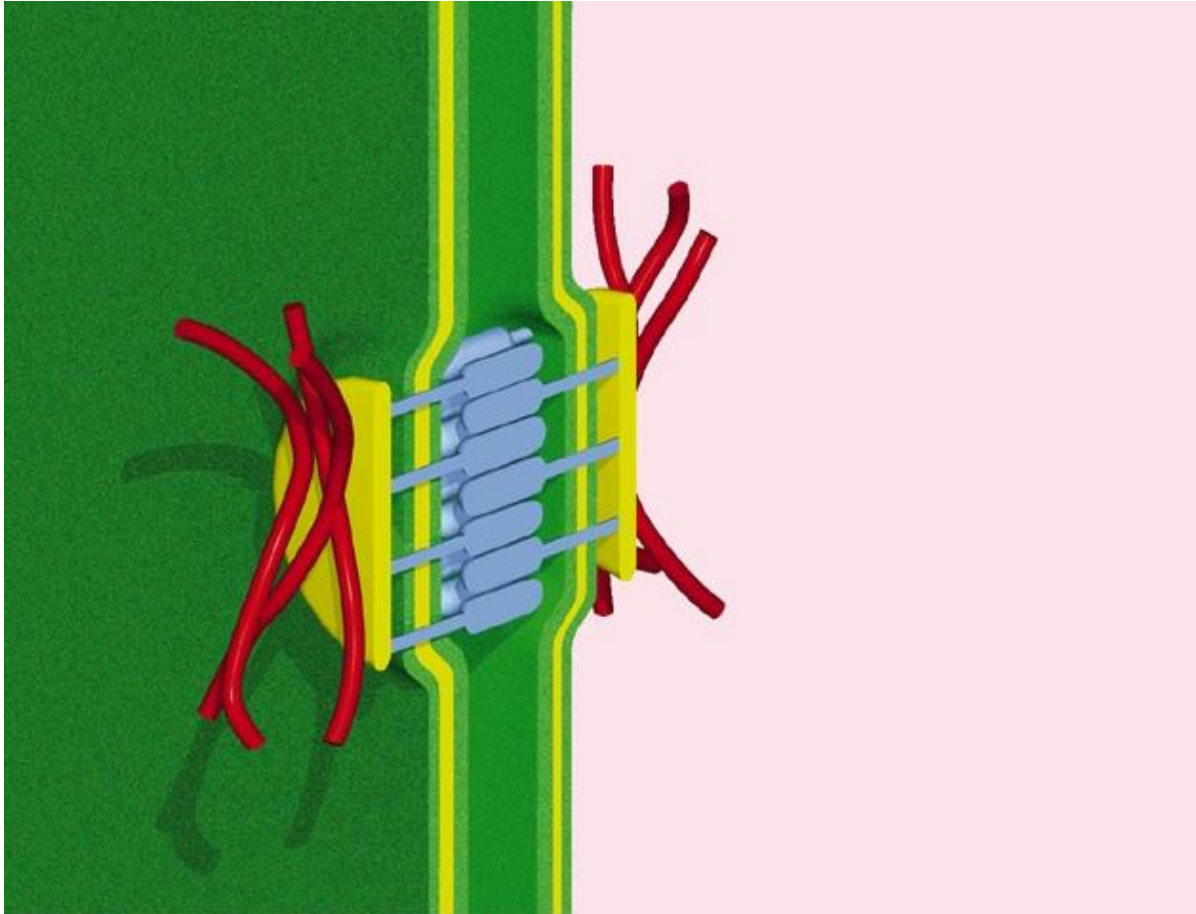


Intercalated discs

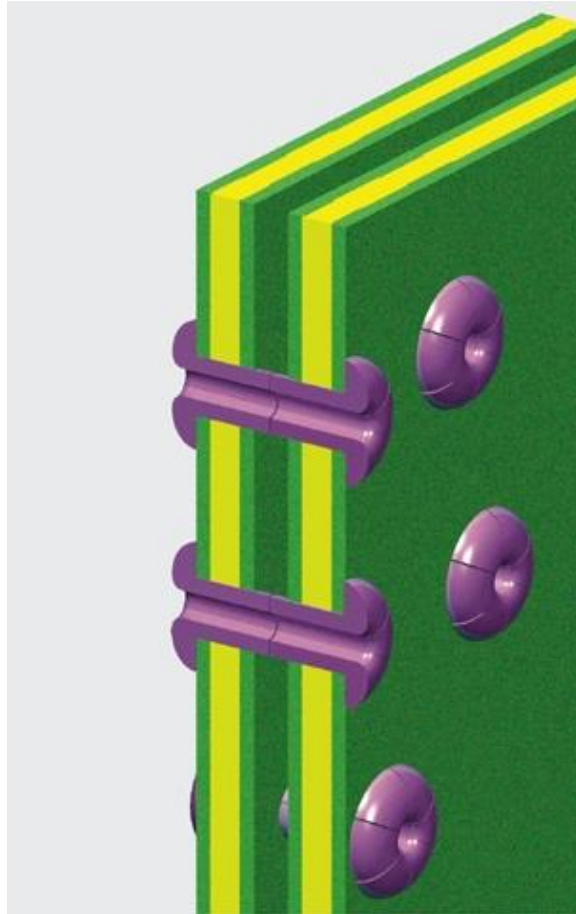
Intercalated discs



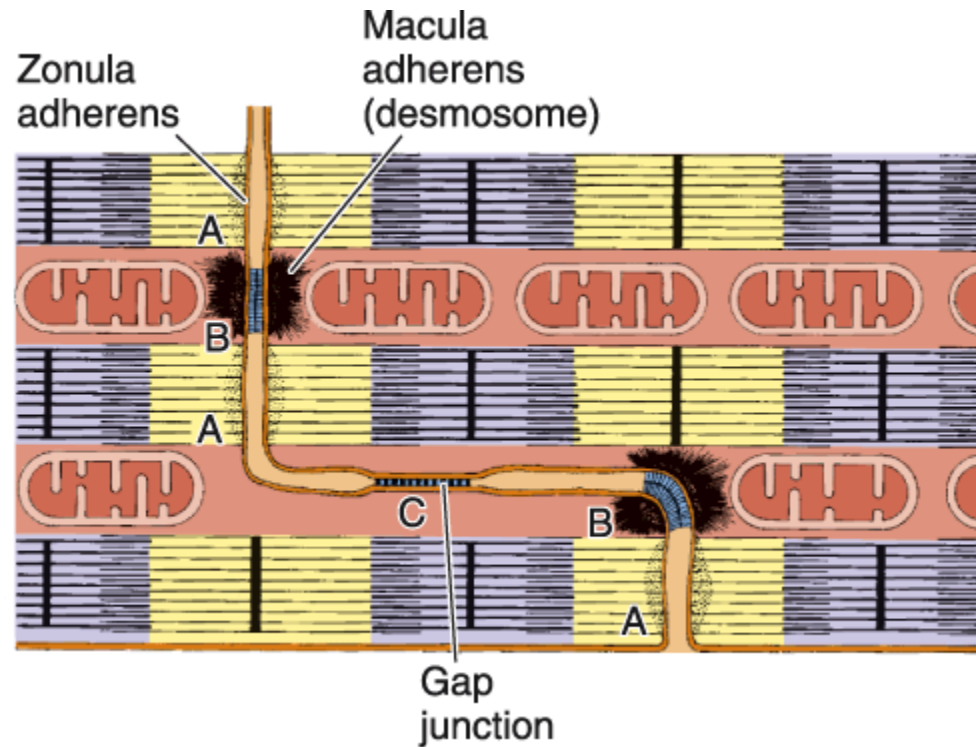
2. Macula adherens (spot desmosome)



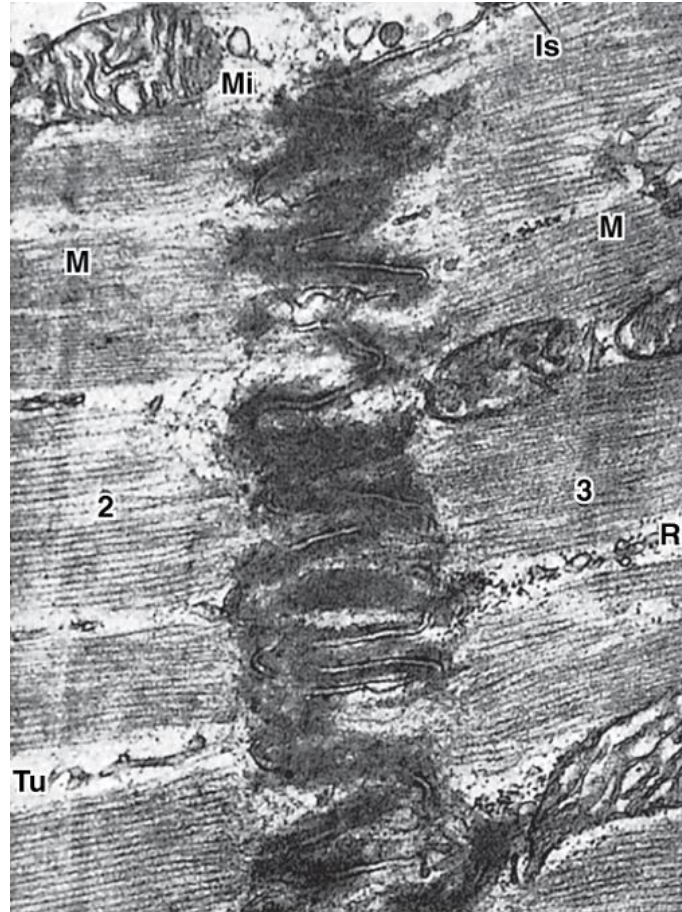
Gap junctions (=communicating=nexus)



Intercalated discs

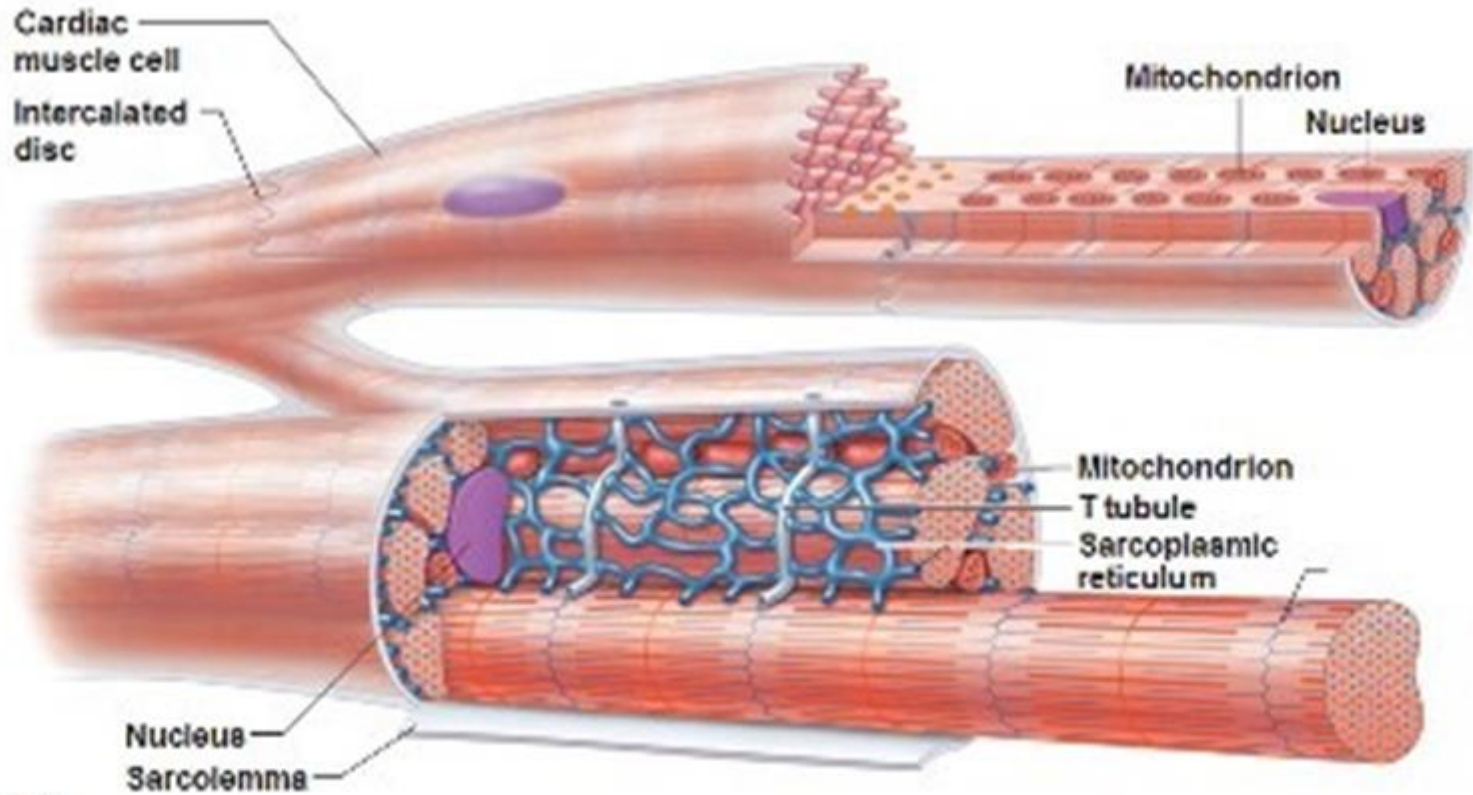
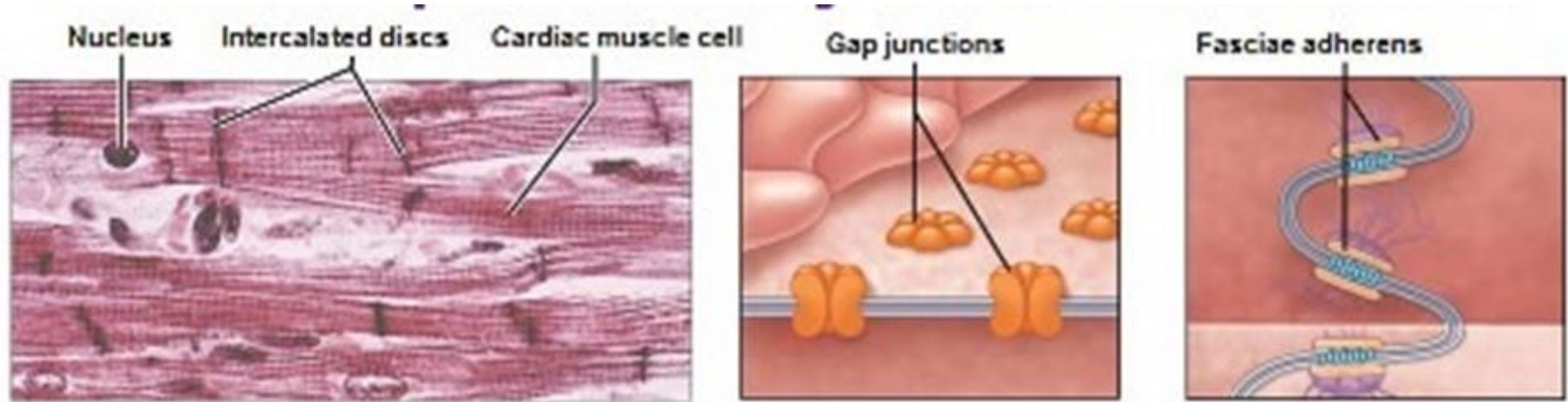


Intercalated discs



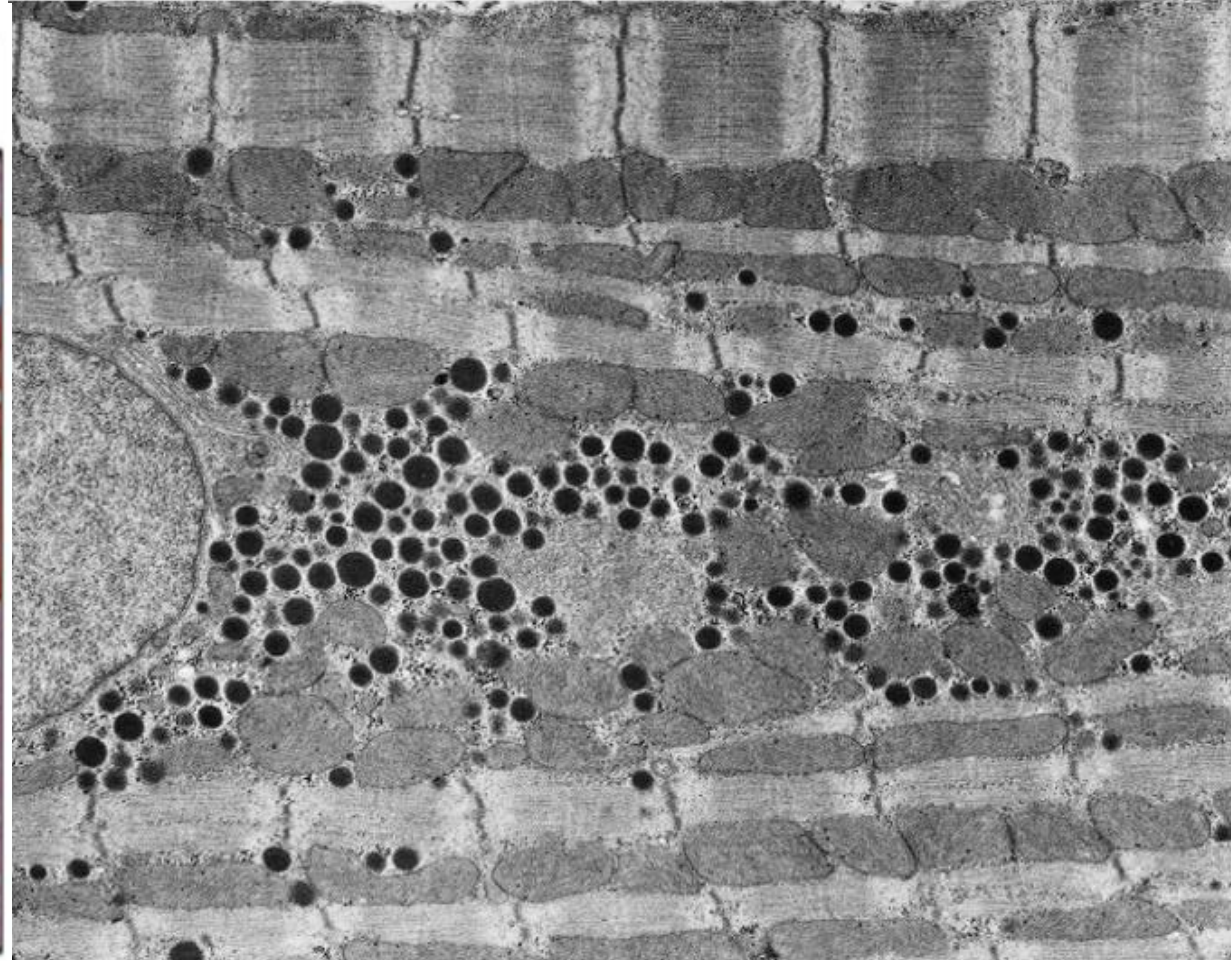
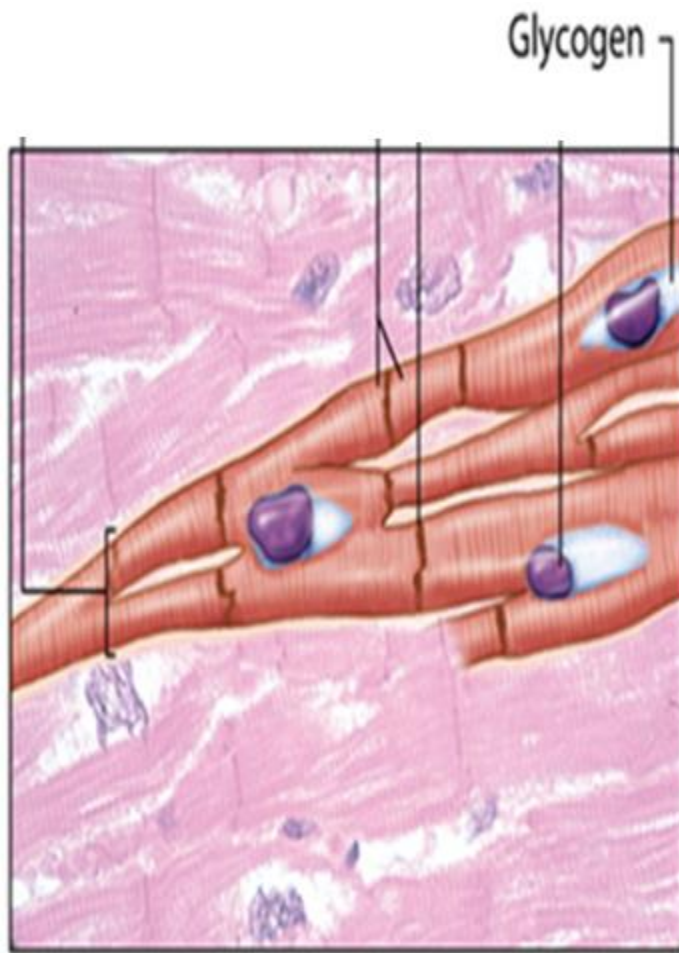
Cardiac muscle

(a)



(b)

Atrial granules



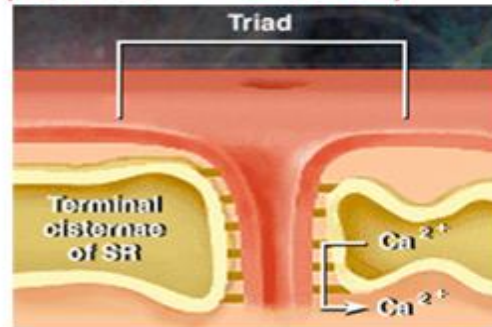
Atrial granules are most abundant in muscle cells of **right atrium** (~600 per cell), but smaller quantities are also found in the left atrium & ventricles.

Skeletal muscle

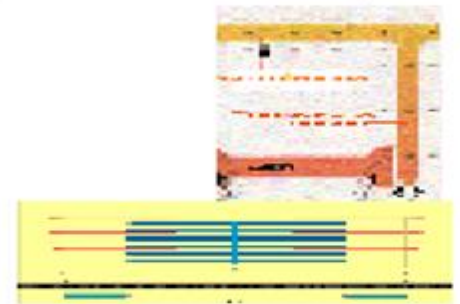
Cardiac muscle

Triad or Diad

Triad at A-I J
(1TT + 2TC)



Diad at Z line
(1TT + 1TC)



Number in 1 sarcomere

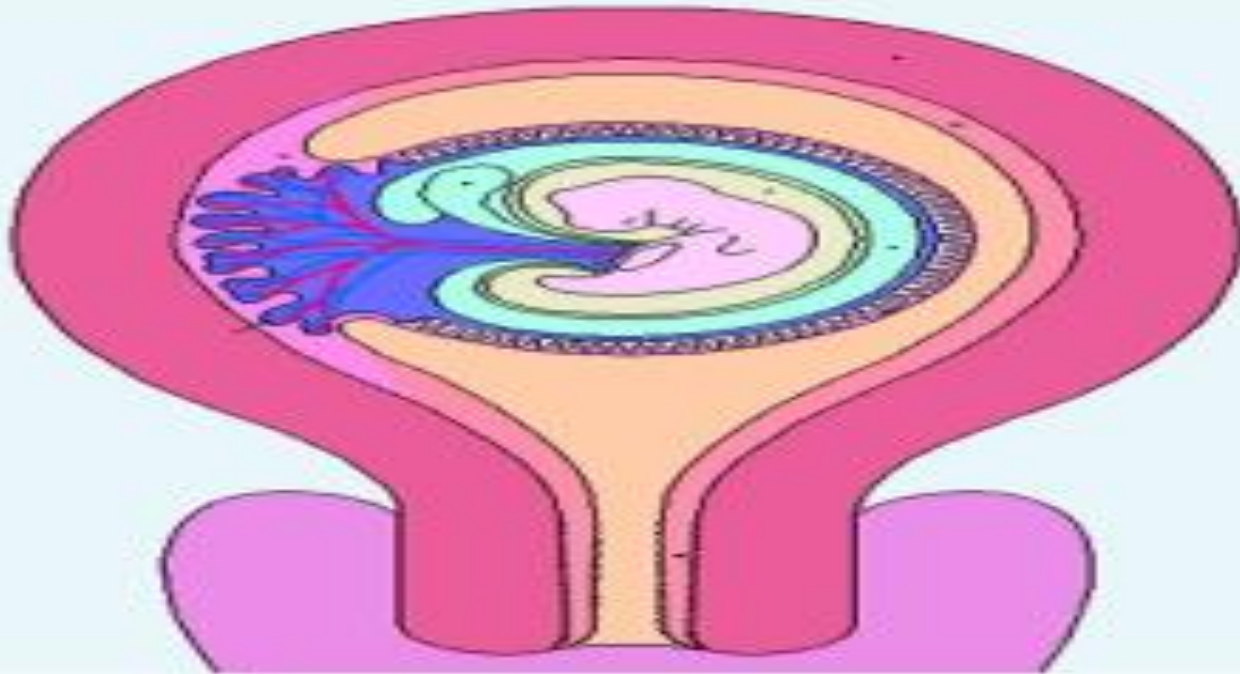
2 Triads

1 Diad

Smooth muscle

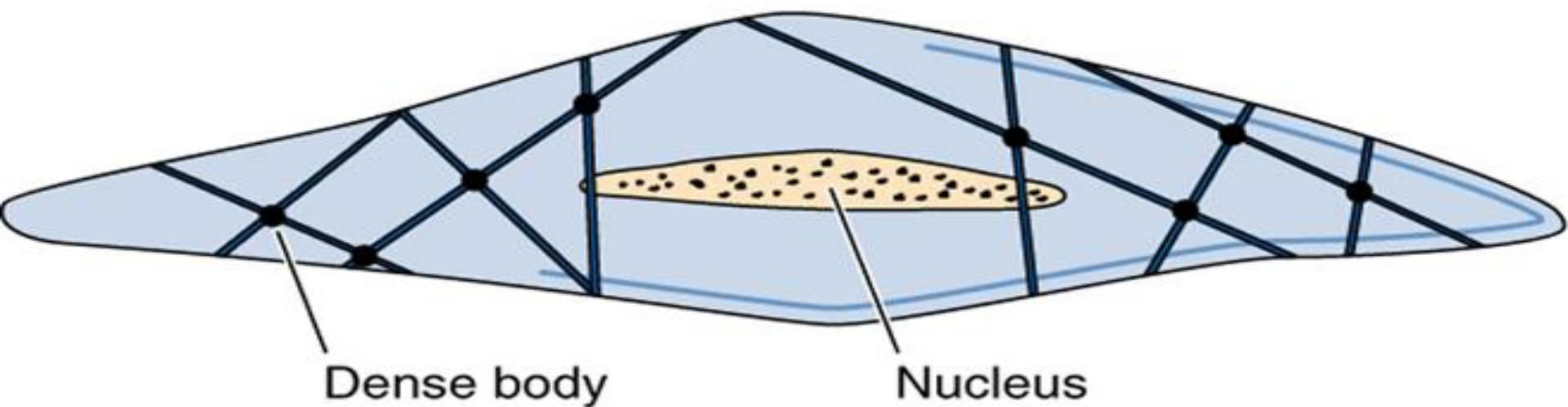
- The **absence of striations** is due to a different organization of contractile filaments.
- Actin and myosin are present but are not arranged in sarcomeres. Instead, thin filaments (actin) are anchored to protein structures called **dense bodies**, which are scattered throughout the sarcoplasm and **attached to the sarcolemma**.
- These dense bodies are functionally analogous to **the Z-discs** of striated muscle.
- The filaments form a **crisscrossing, lattice-like network**. When the cell contracts, this network **pulls on the dense bodies**, causing the entire cell to **shorten** and "ball up."
- The sarcolemma of smooth muscle cells has **small invaginations** called caveolae, which are believed to function **similarly to the T-tubule system** in **regulating calcium influx for contraction**.

Smooth muscle

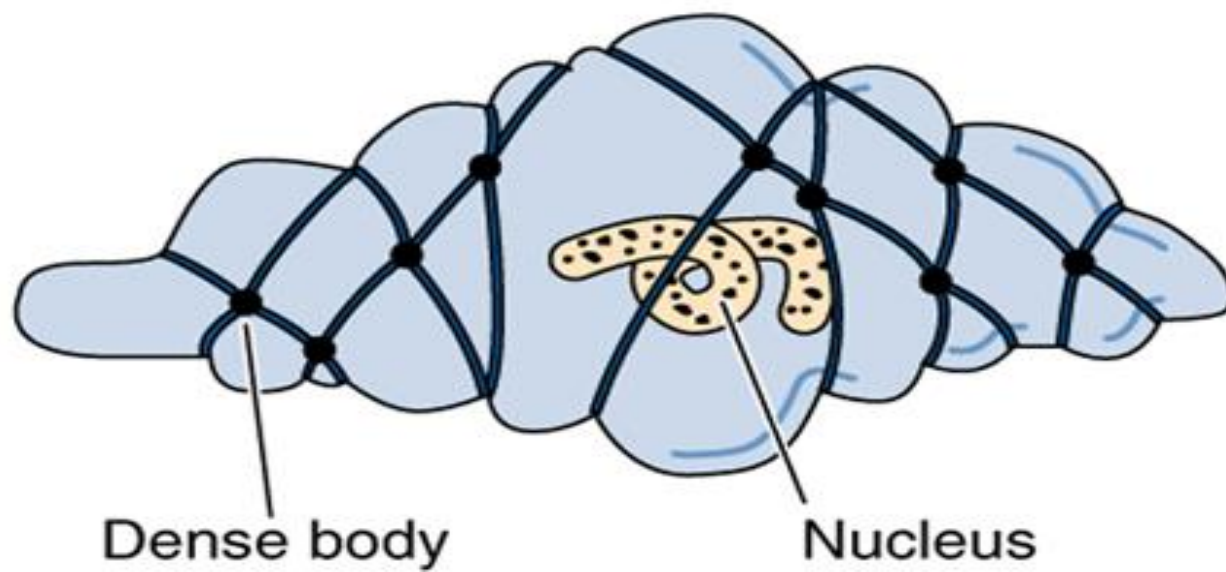


- **Hypertrophy** is an increase of cell volume.
- **Hyperplasia** is an increase in the number.

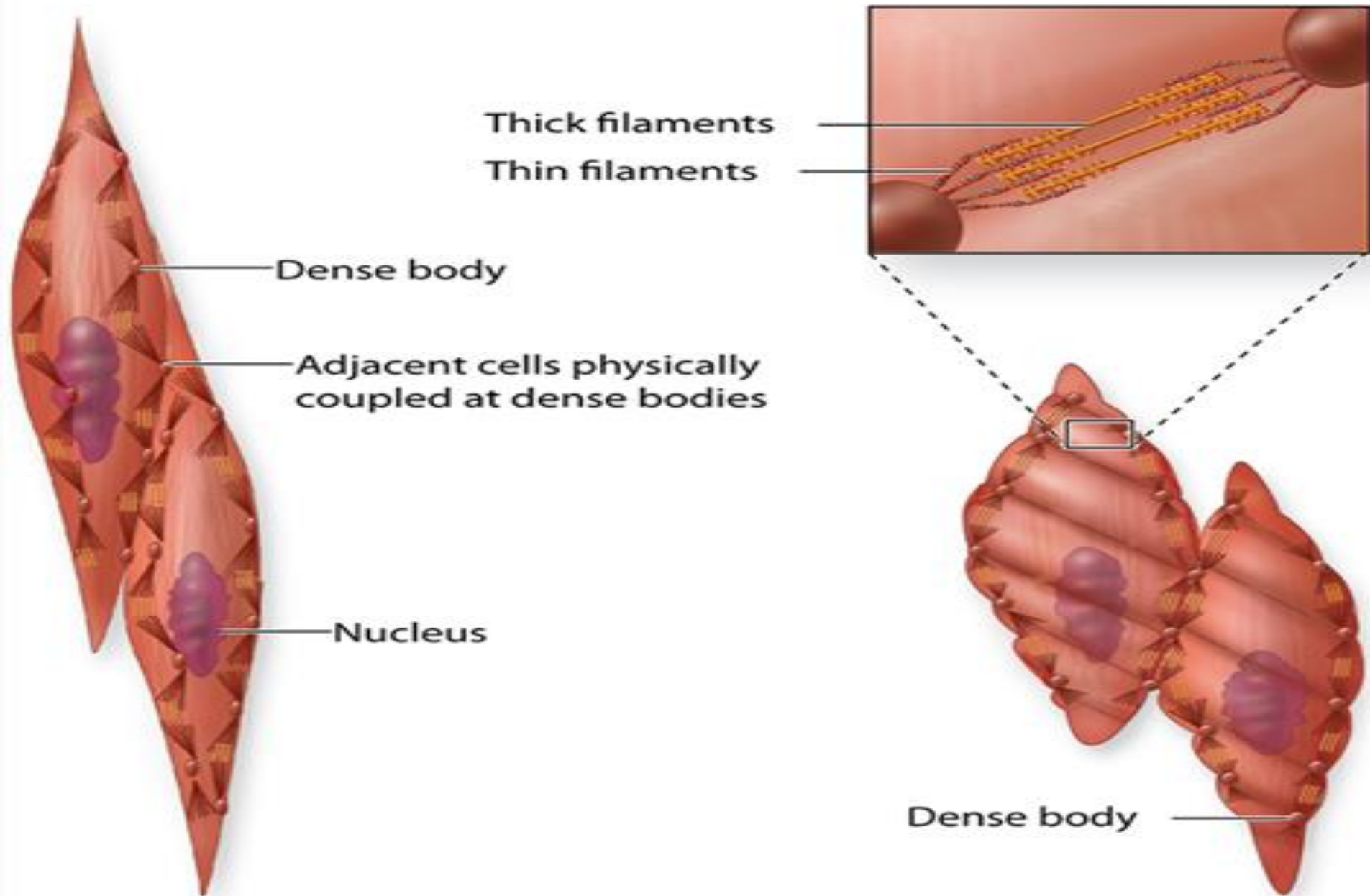
Relaxed smooth muscle cell



Contracted smooth muscle cell



The contractile mechanism of smooth muscle



Muscle Characteristics (VIP)

Features	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Striations	Yes	Yes	No
Fibers	Long, cylindrical, unbranched	Short, branched, anastomosing	Short, spindle shaped
Nuclei	Multiple, peripheral in cell	Single, central in cell	Single, central in cell
Cell junctions	No	Intercalated disks	Gap (nexus) junctions
T tubules	Well developed	Well developed	No
Sarcoplasmic reticulum	Highly developed; has terminal cisterns	Less well developed; small cisterns	Present, but poorly developed
Regeneration	Yes, satellite cells	No	Yes, mitosis
Contraction	Initiated by nerve action potential	Spontaneous; pacemaker system; modulated by nervous system and hormones	Spontaneous; modulated by nervous system and hormones
Main function	Voluntary movement of limbs, digits, face, tongue, and other muscles	Involuntary rhythmic contractions; pumps blood to muscles and organs; modulated by physiological and emotional factors	Involuntary control of blood vessel diameter, gut peristalsis, uterine contractions during childbirth, airway diameter, and others

Feature	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Key Histological Feature	Sarcomere organization; peripheral nuclei	Intercalated discs; branched cells	Fusiform shape; lack of striations; dense bodies
Cell-Cell Junctions	None; fibers are independent	Intercalated discs (gap junctions, desmosomes, fascia adherens)	Gap junctions
T-Tubules & SR	Well-developed; forms triads (1 T-tubule, 2 terminal cisternae)	Present; forms diads (1 T-tubule, 1 terminal cisterna)	Absent